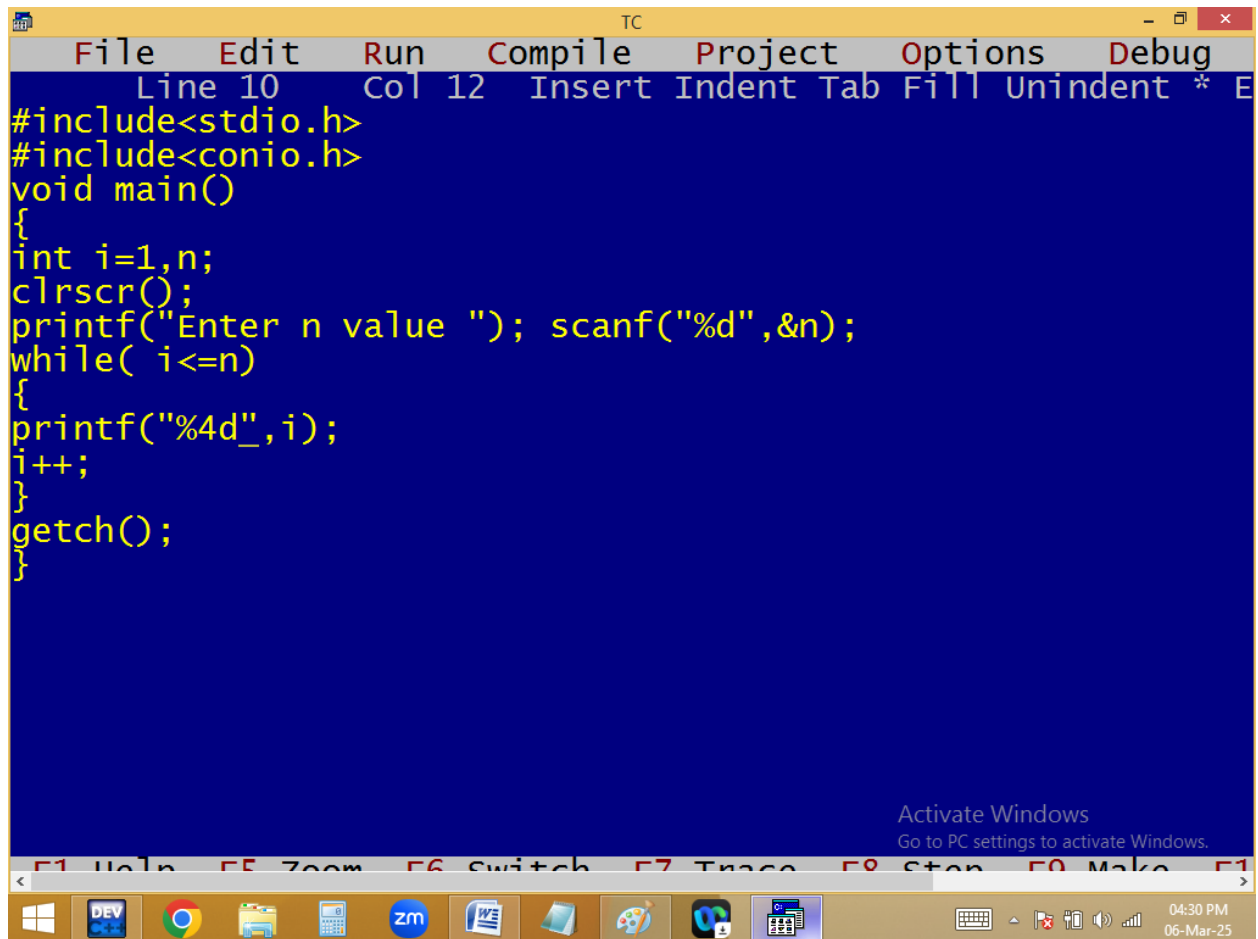


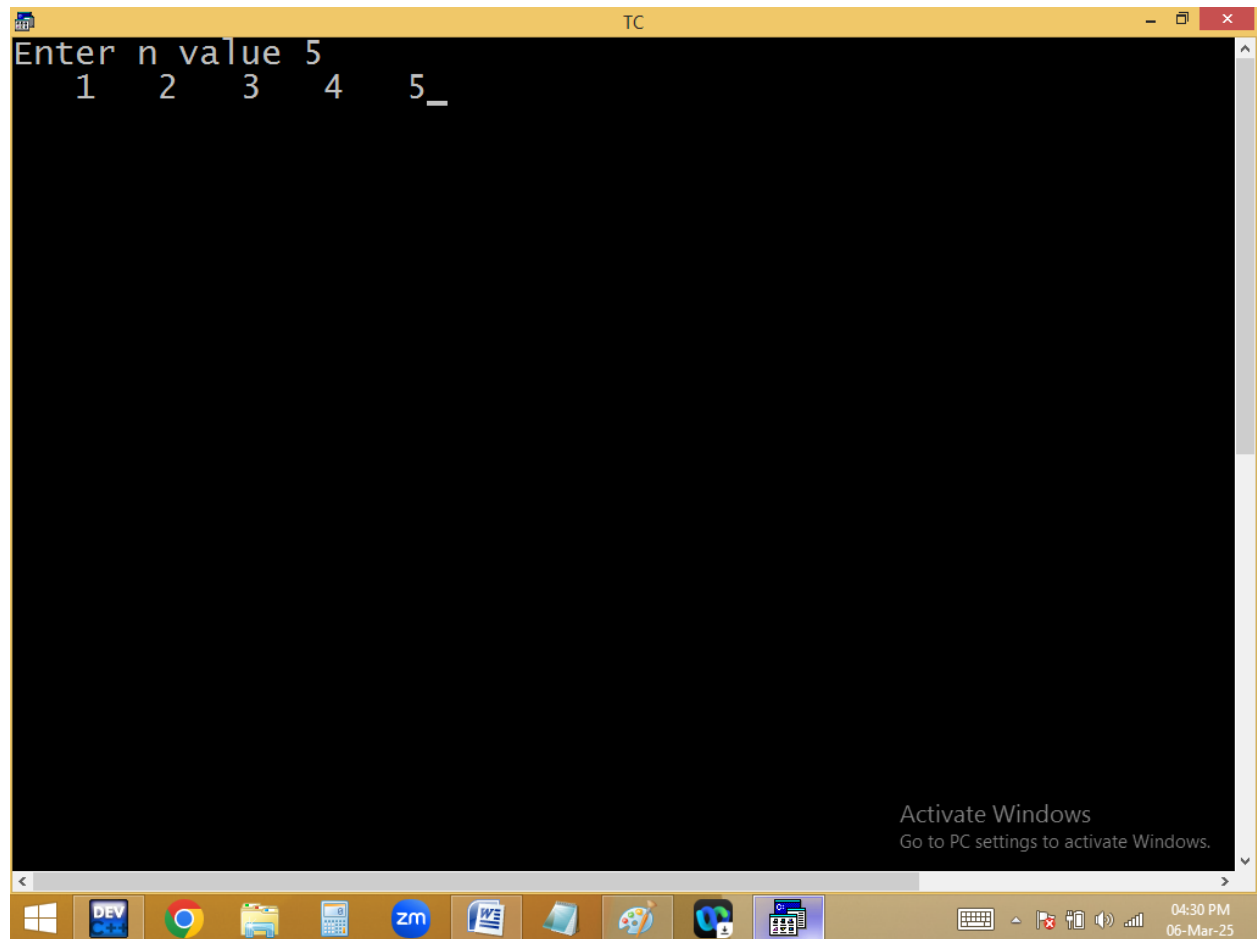
Printing 1..n natural no's using while loop:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 12" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while( i<=n)
{
printf("%4d_",i);
i++;
}
getch();
}
```

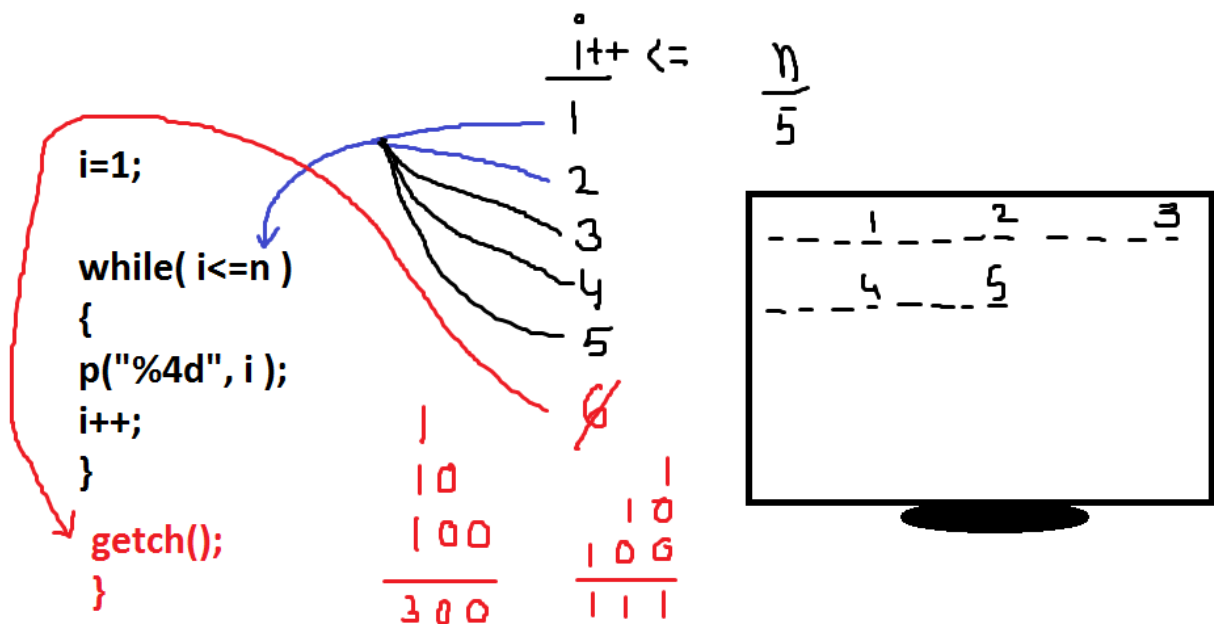
At the bottom of the window, there is a taskbar with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as 04:30 PM and the date as 06-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

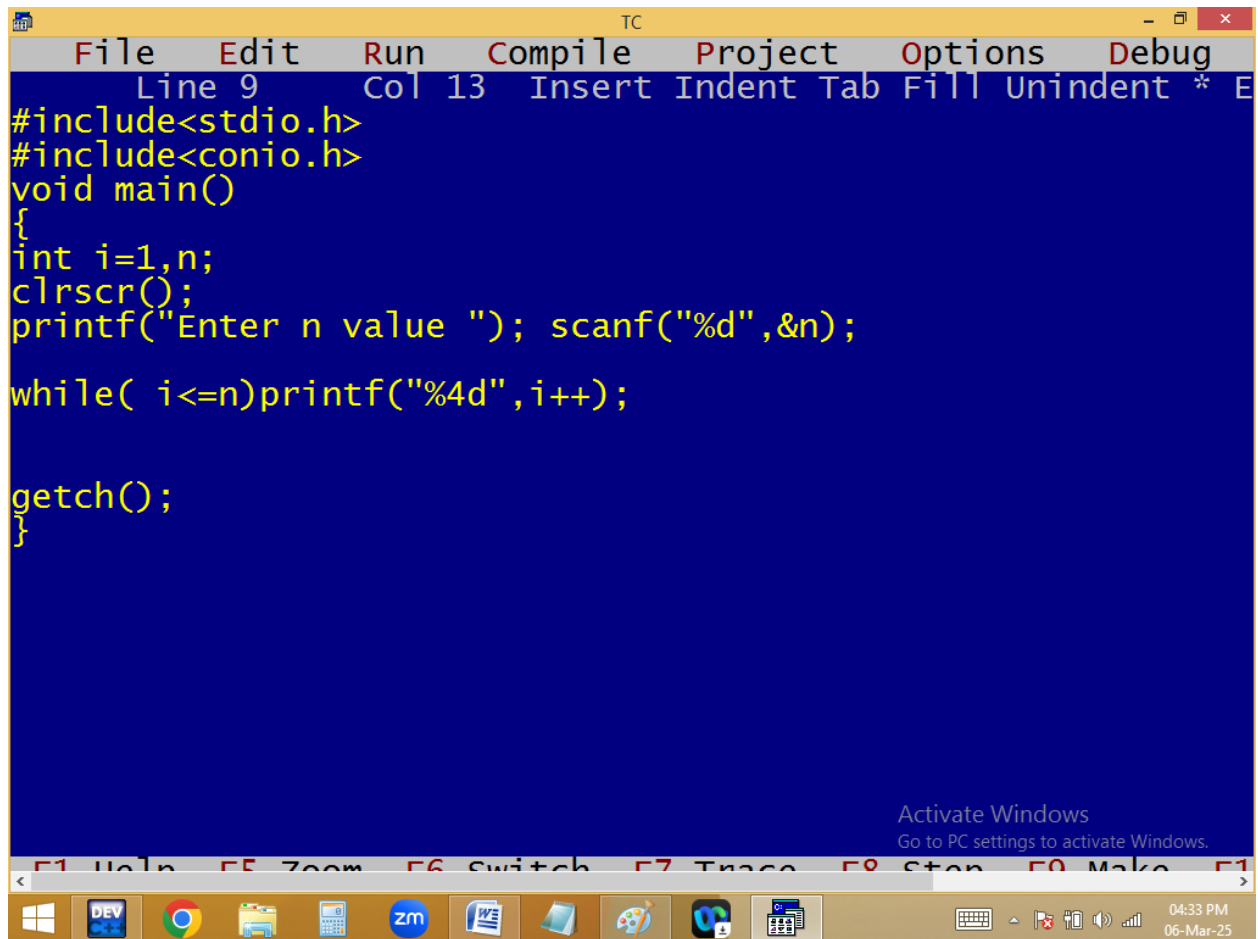


```

Enter n value 200
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135
141 142 143 144 145 146 147 148 149 150 151 152 153 154 155
161 162 163 164 165 166 167 168 169 170 171 172 173 174 175
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195

```





```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);

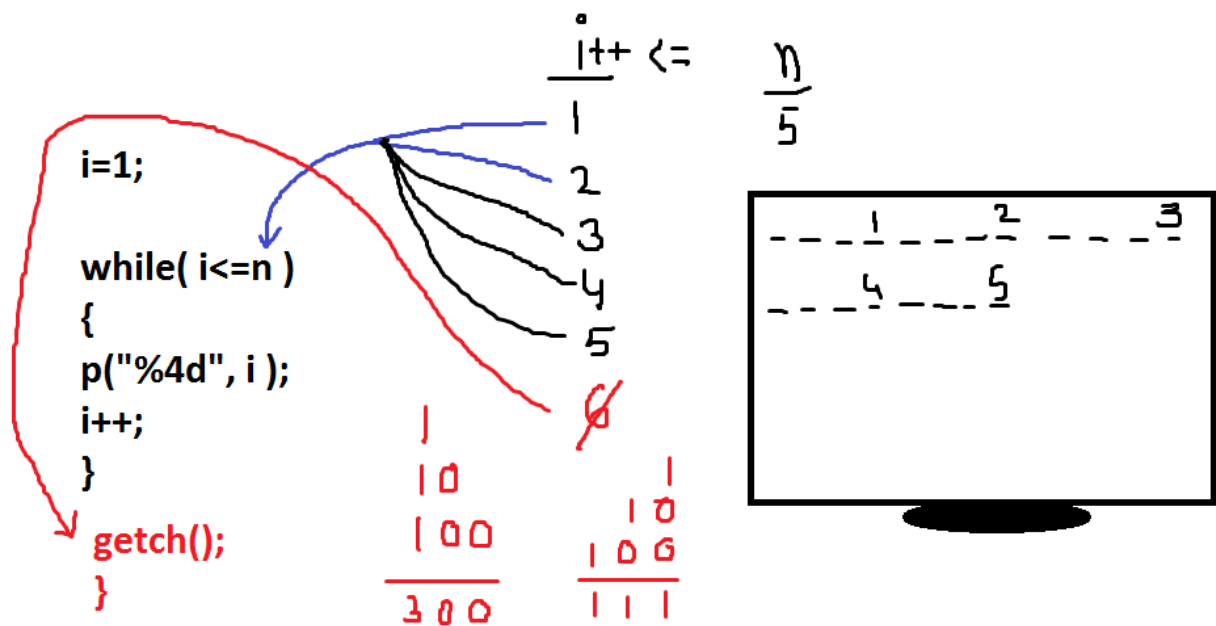
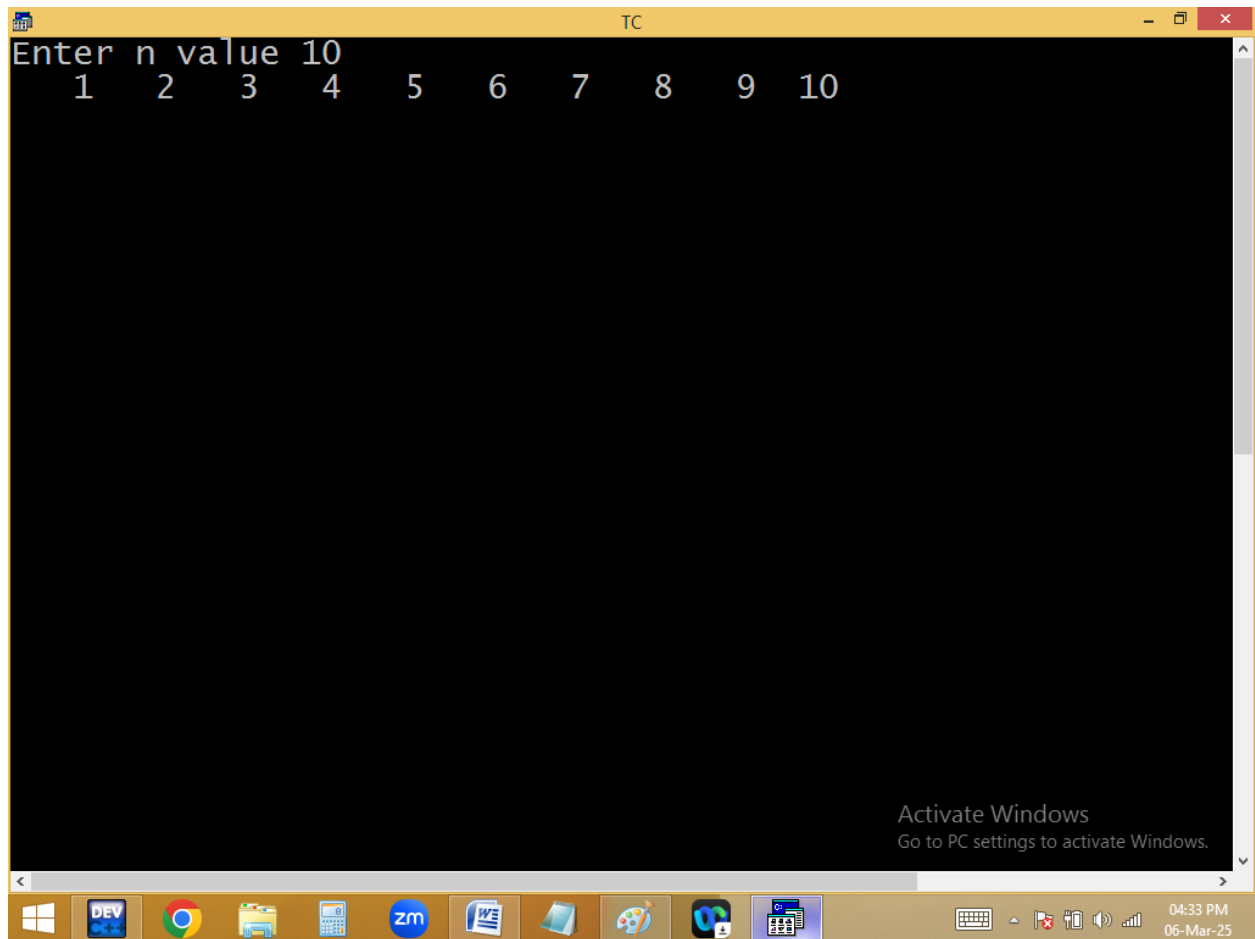
while( i<=n)printf("%4d",i++);

getch();
}
```

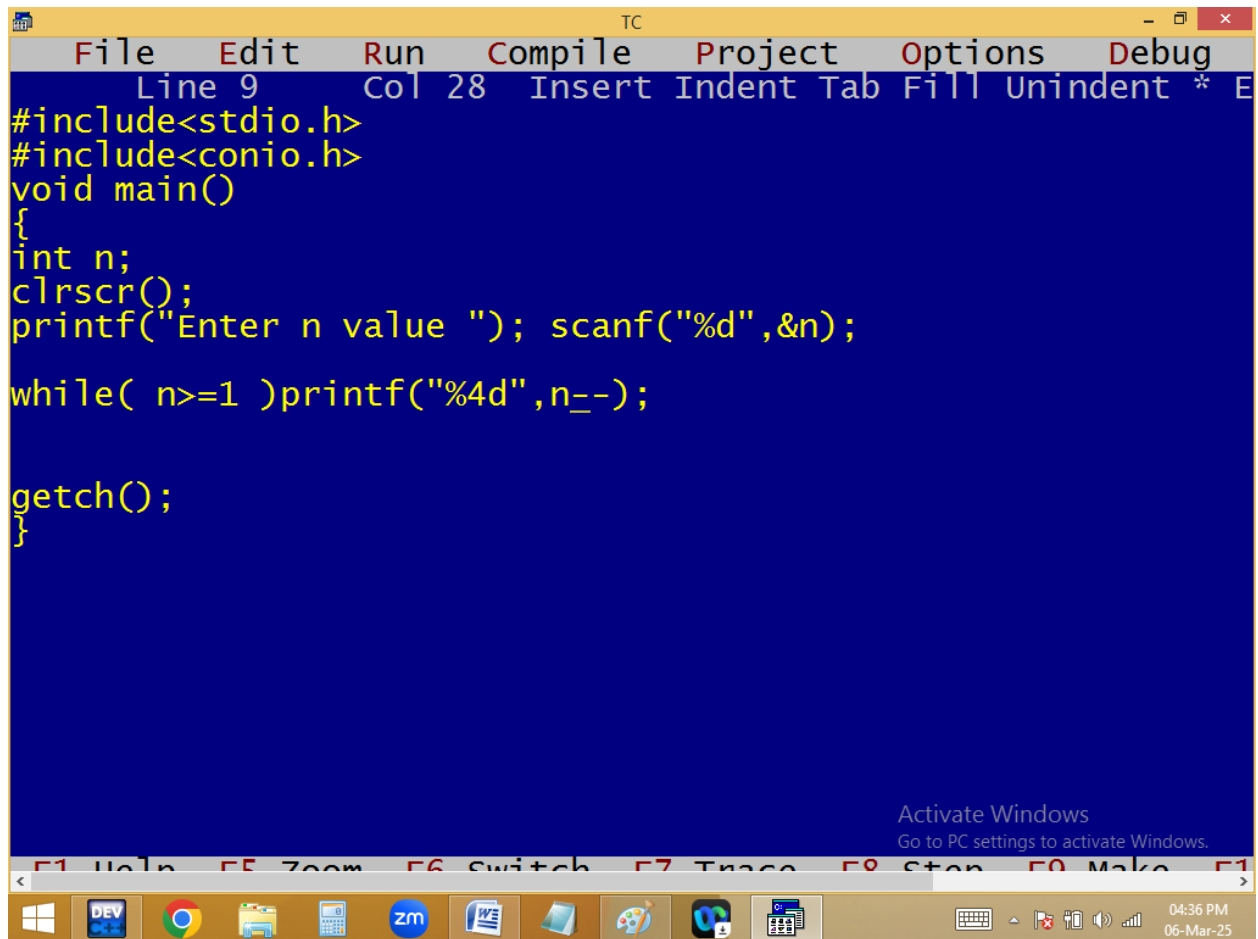
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:33 PM
06-Mar-25



Print 1..n in reverse order:



```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 28 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);

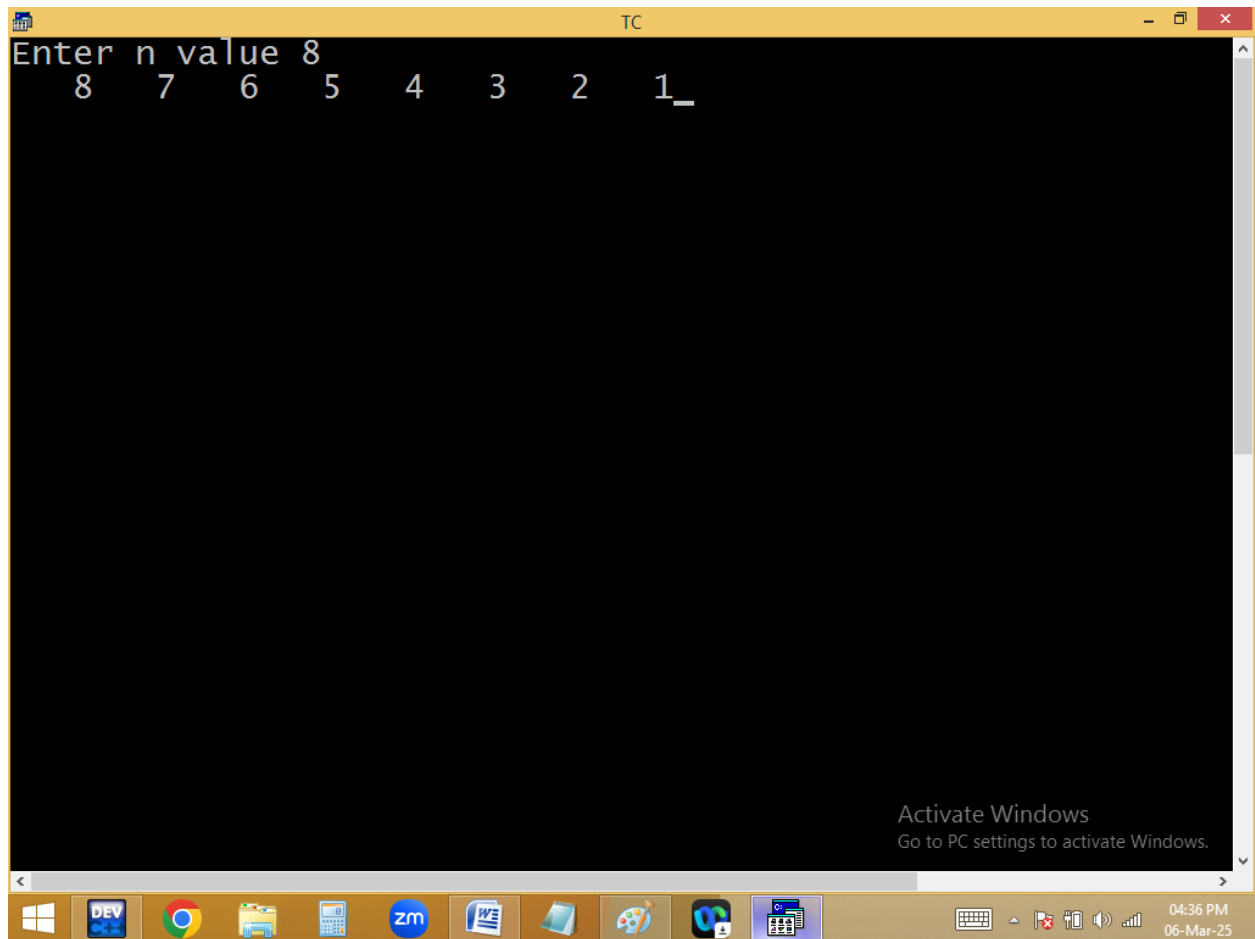
while( n>=1 )printf("%4d",n_);

getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:36 PM
06-Mar-25



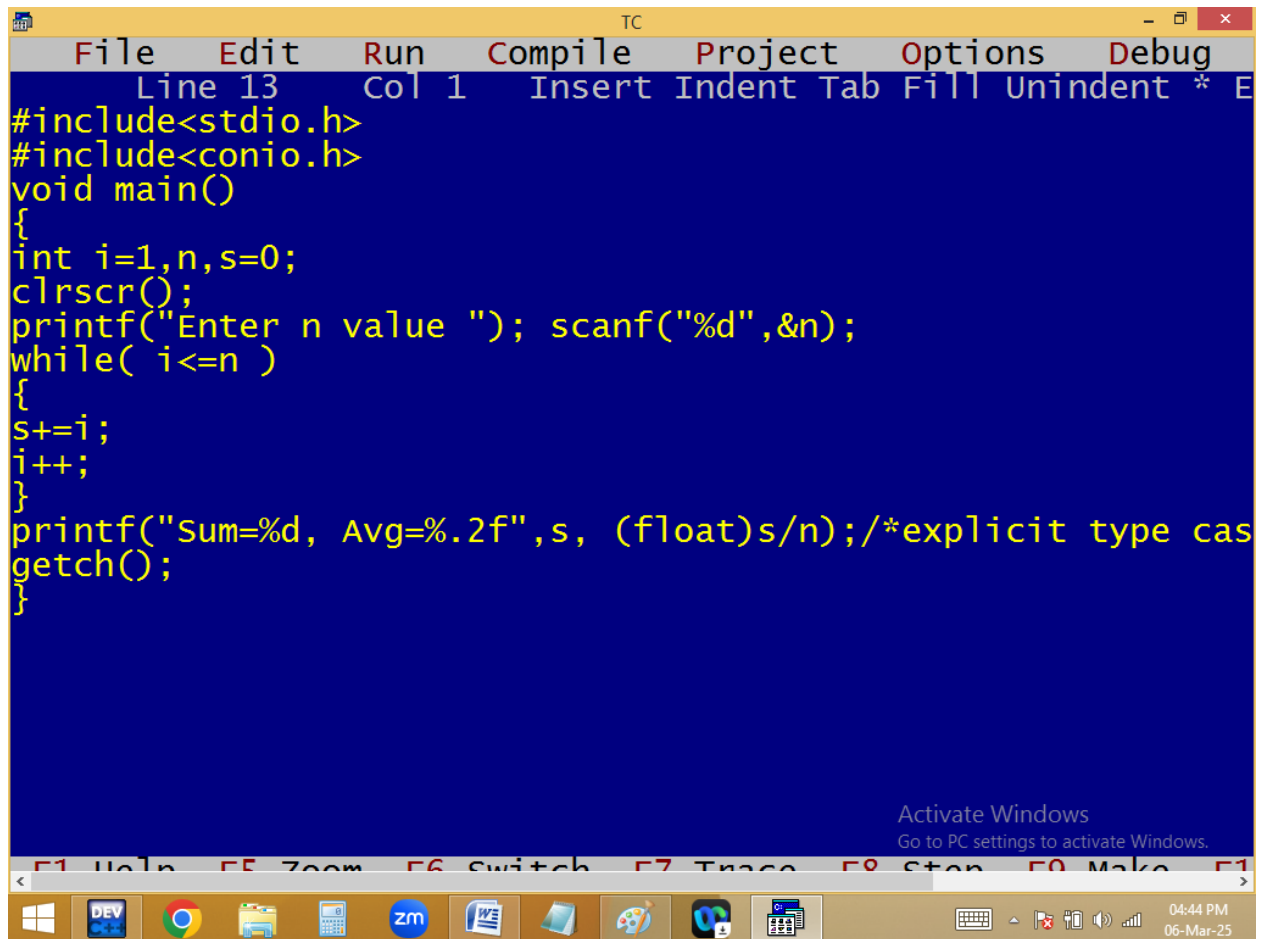
The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text displays the prompt "Enter n value 8" followed by a sequence of numbers from 8 down to 1, each on a new line. The window title bar reads "TC". The Windows taskbar at the bottom includes icons for various applications and the system clock showing 04:36 PM on 06-Mar-25.

```
Enter n value 8
8
7
6
5
4
3
2
1_
```

Print 1..n numbers sum and avg[mean]:

For example 1..4 numbers sum is $1+2+3+4=10$

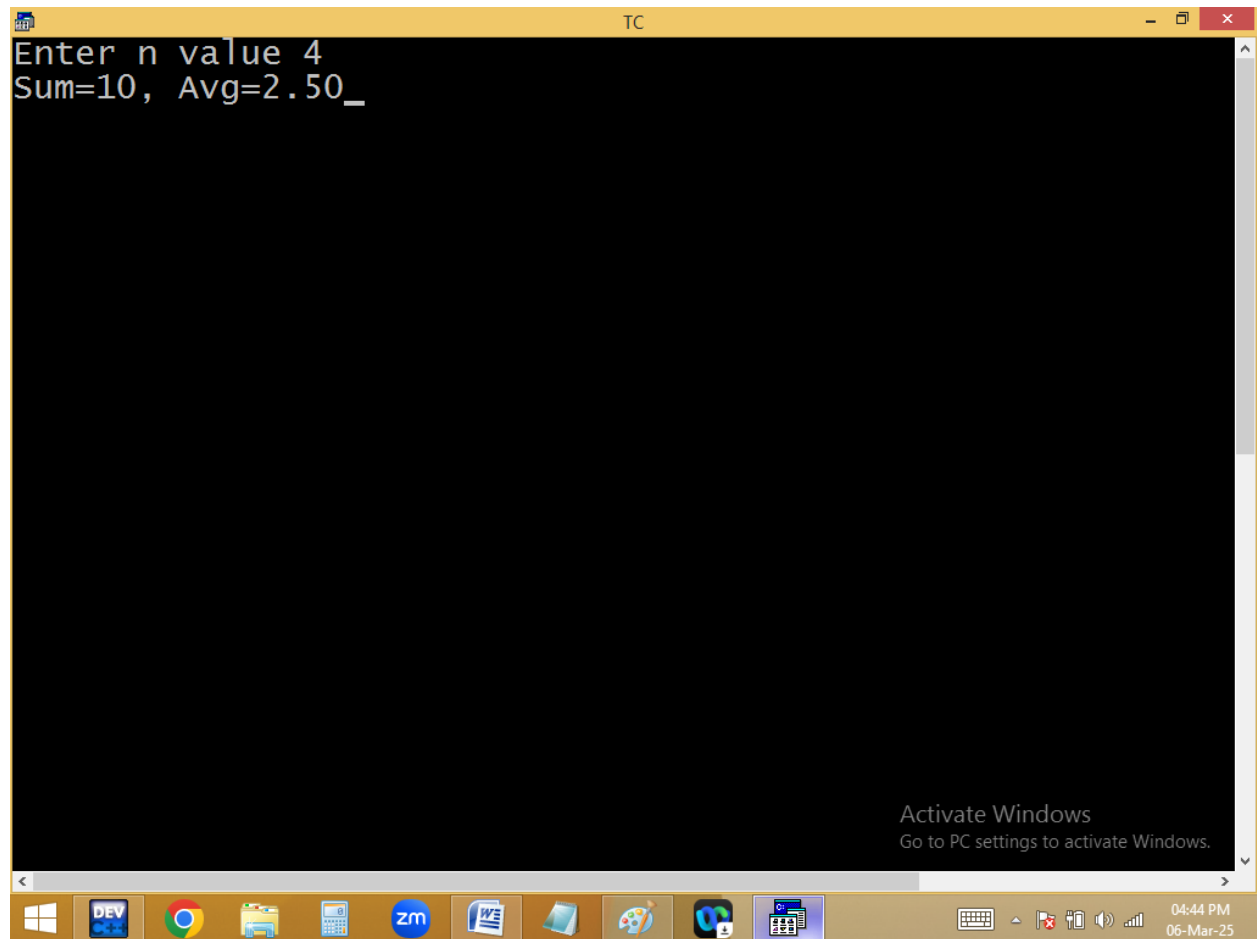
and $\text{avg}=2.5$

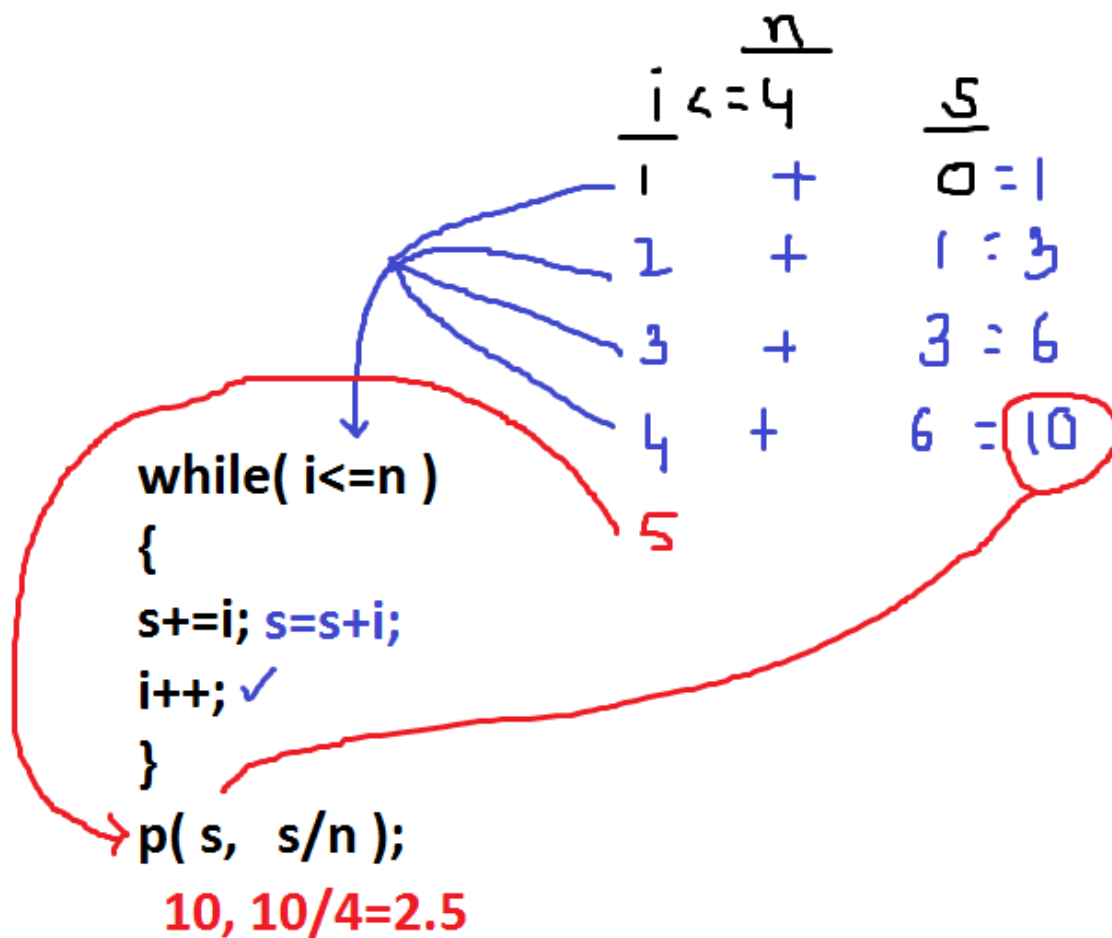


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains the following C code:

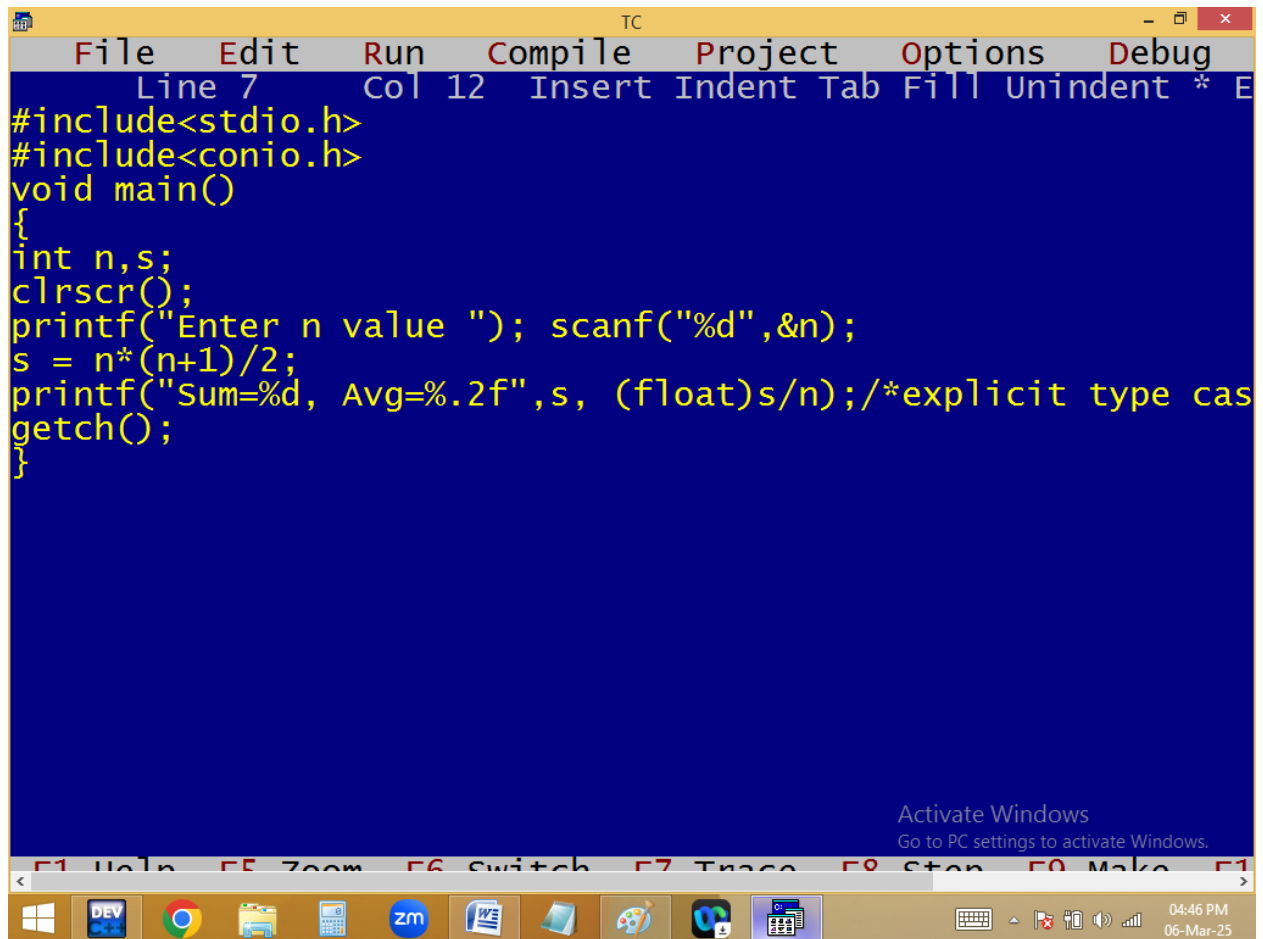
```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while( i<=n )
{
s+=i;
i++;
}
printf("Sum=%d, Avg=%.2f",s, (float)s/n);/*explicit type cas
getch();
}
```

At the bottom of the window, there is a taskbar with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. A system tray on the right shows the time as 04:44 PM and the date as 06-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





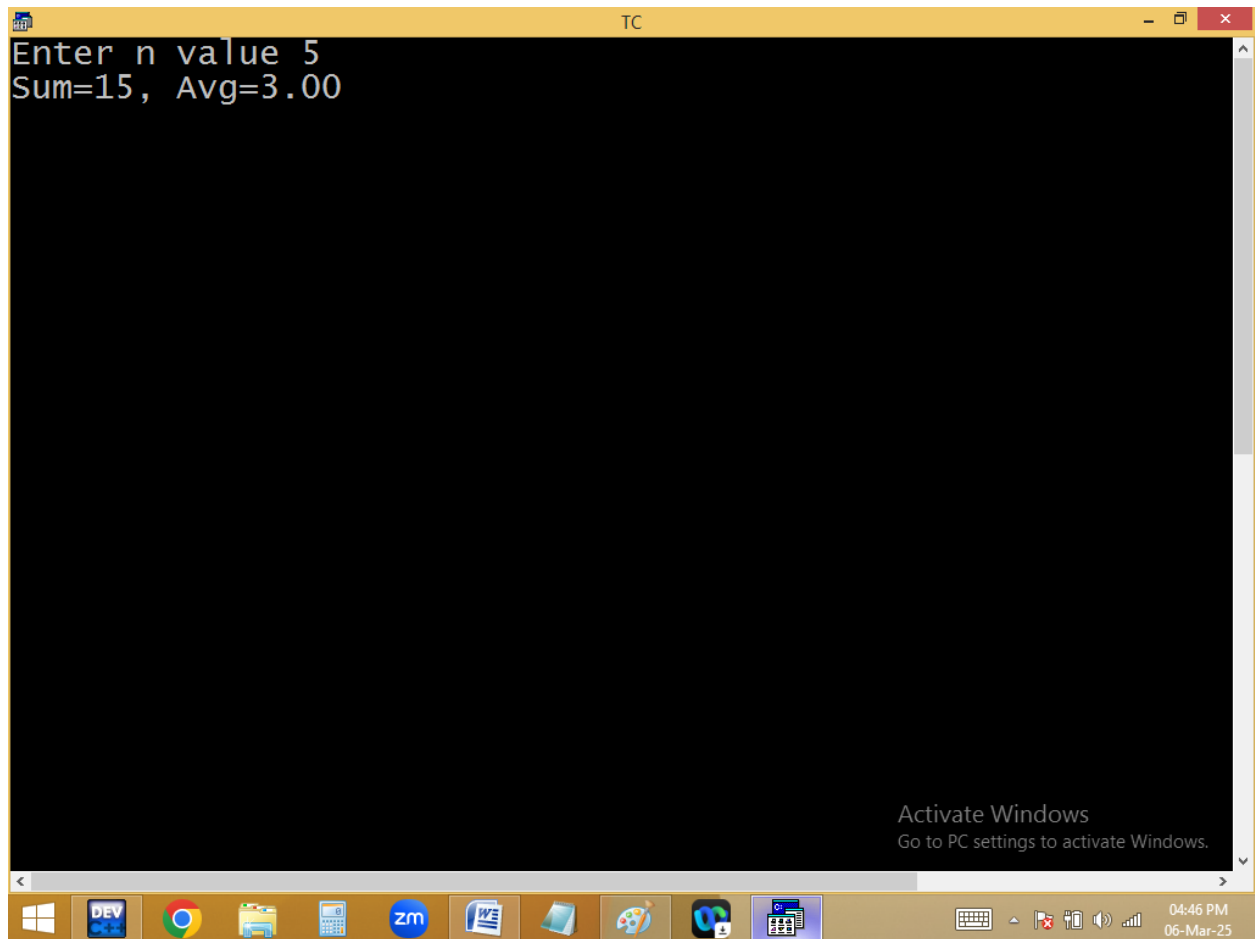
Without using loop/goto label:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 7", "Col 12", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,s;
clrscr();
printf("Enter n value "); scanf("%d",&n);
s = n*(n+1)/2;
printf("Sum=%d, Avg=%.2f",s, (float)s/n);/*explicit type cast
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and others. A system tray on the right shows the time as 04:46 PM and the date as 06-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

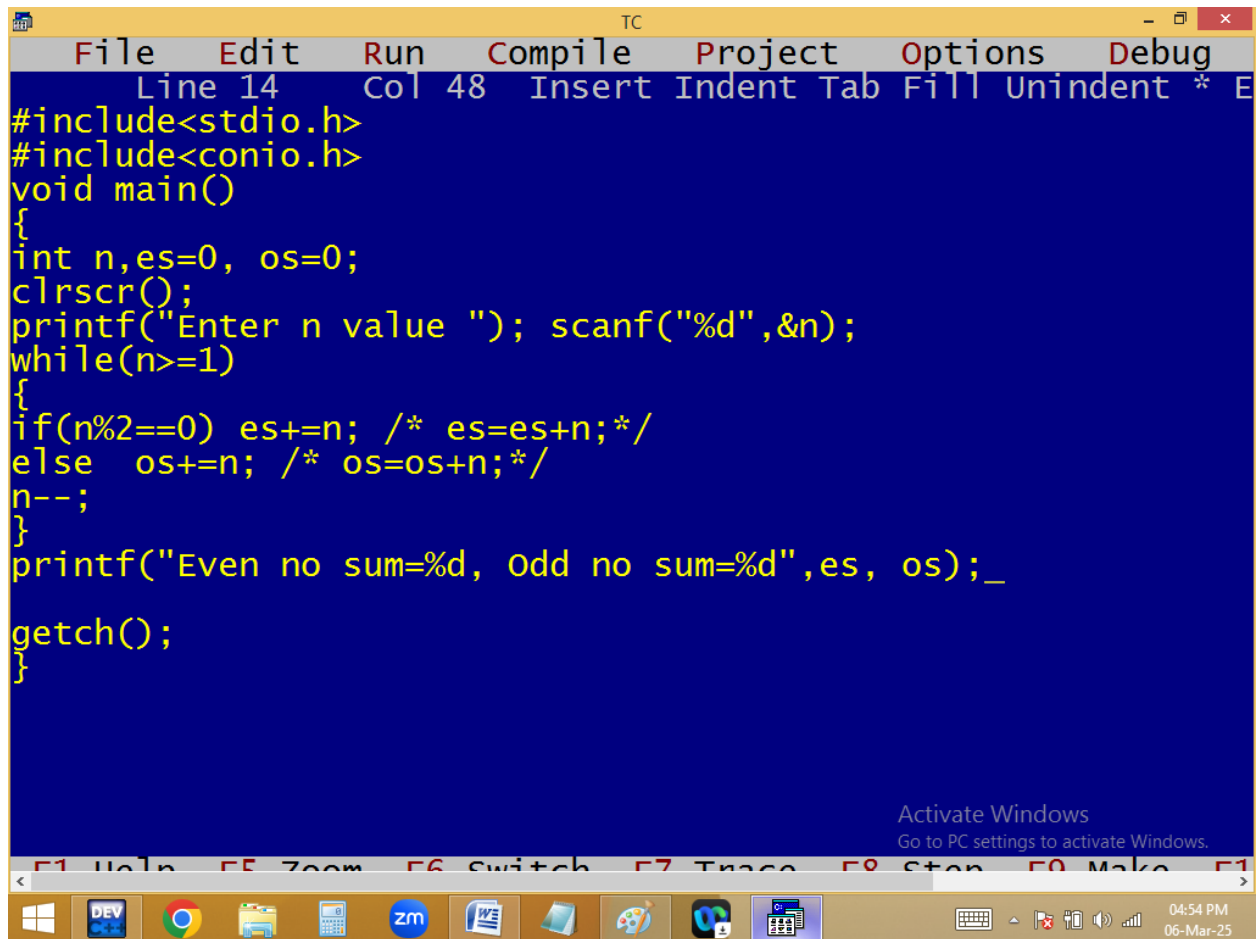


```
Enter n value 5
Sum=15, Avg=3.00
```

$$s = n * (n+1) / 2;$$
$$s = 4 * 5 / 2 = 10$$

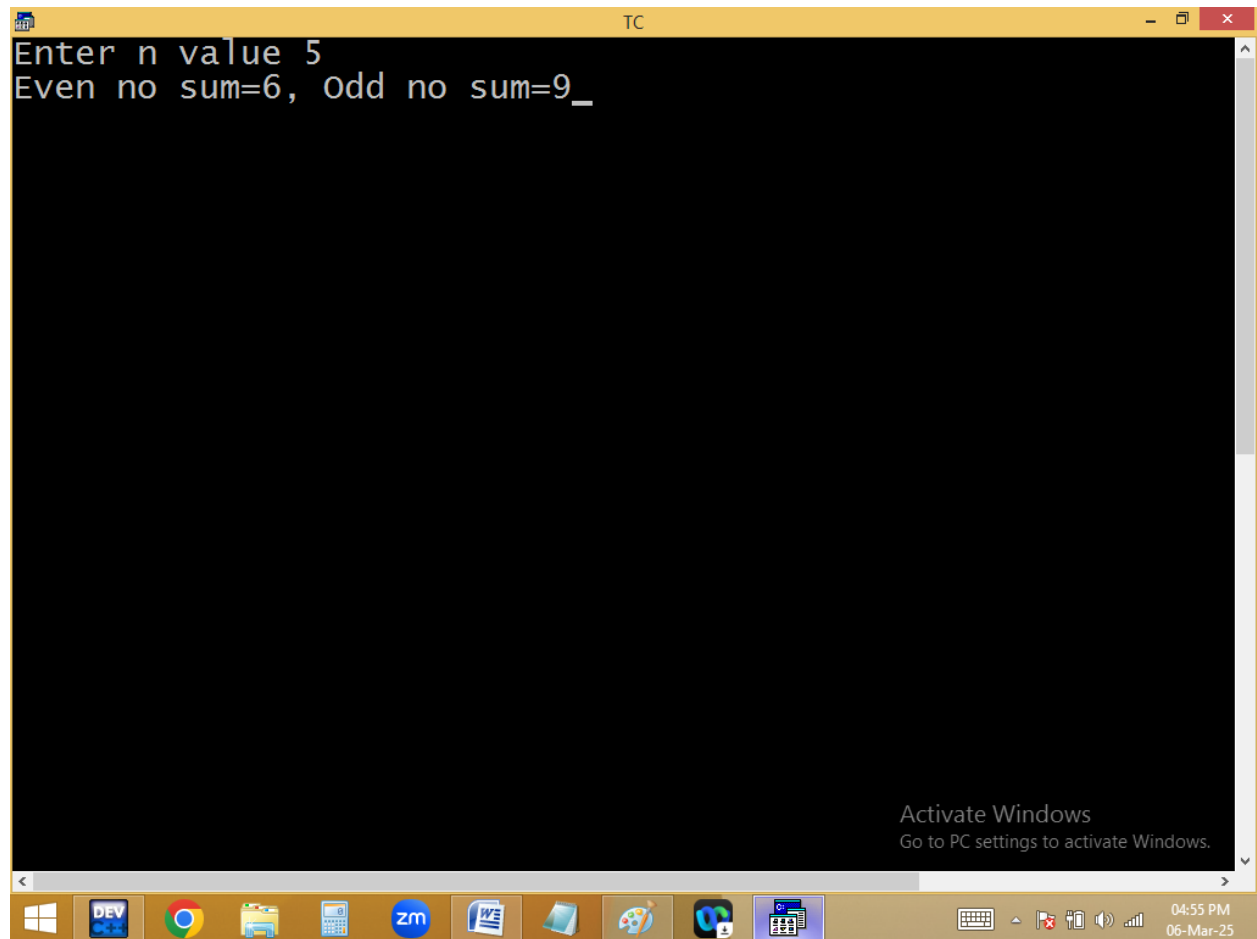
Finding 1..n even , odd no's sum?

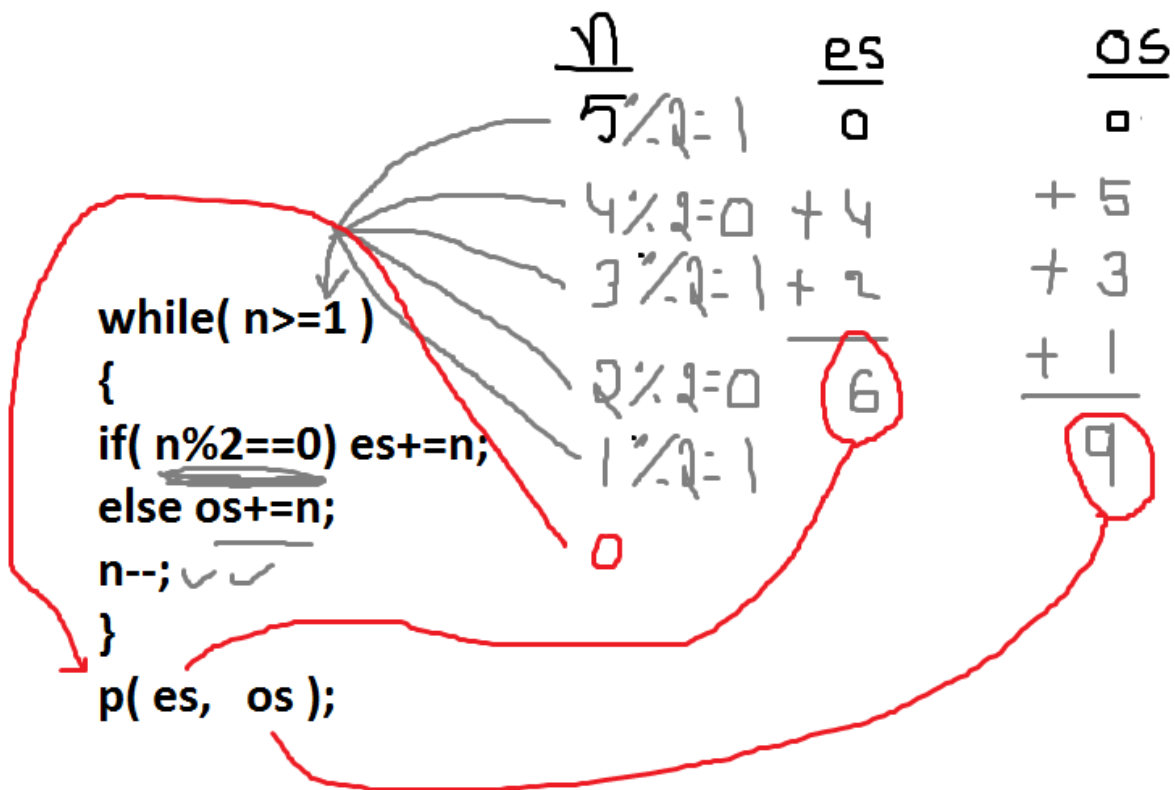
$$n=5 \begin{cases} 2 + 4 = 6 \\ 1 + 3 + 5 = 9 \end{cases}$$



```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 48 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,es=0, os=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
if(n%2==0) es+=n; /* es=es+n;*/
else os+=n; /* os=os+n;*/
n--;
}
printf("Even no sum=%d, Odd no sum=%d",es, os);_
getch();
}

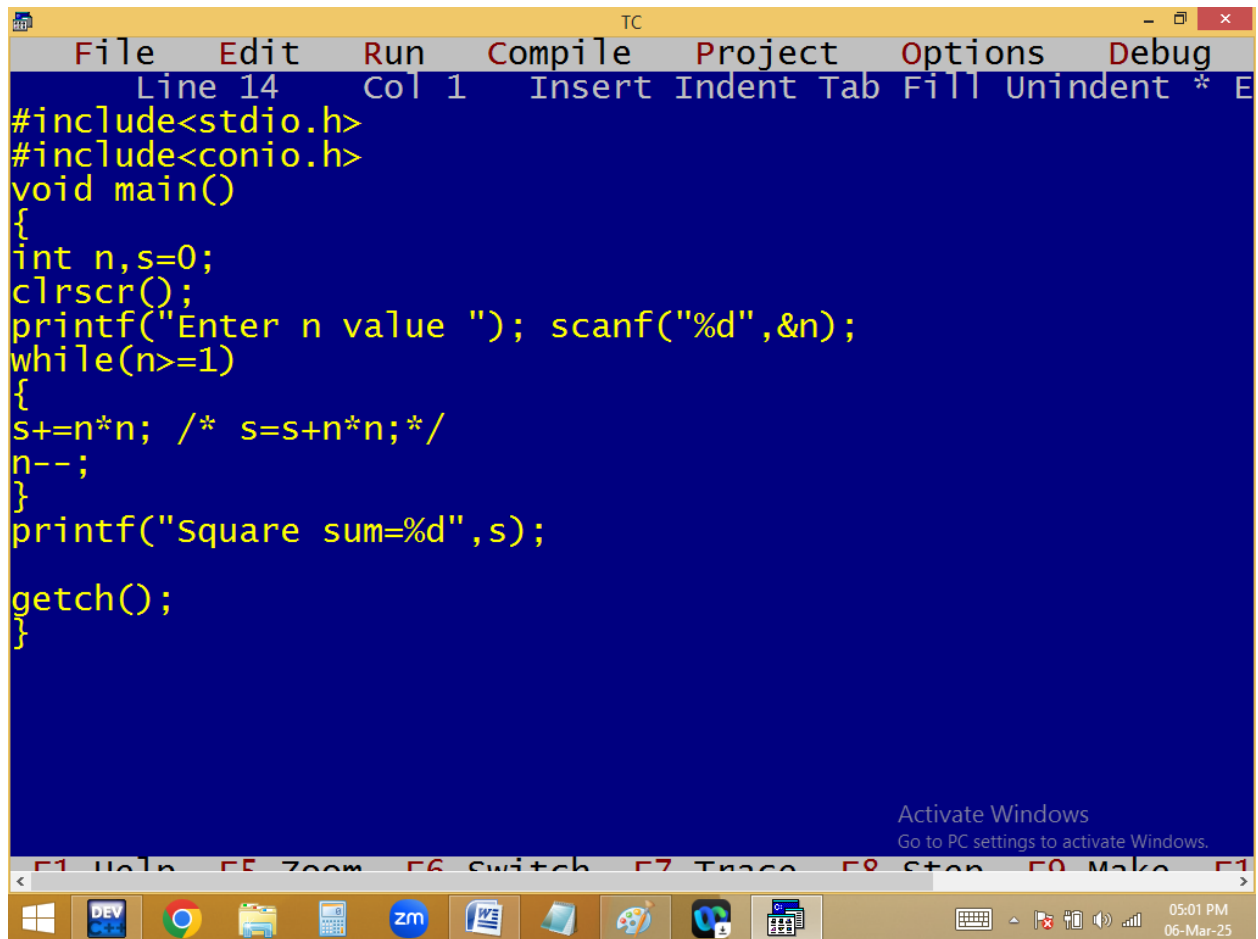
Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```





Write a c program to find 1..n squares sum:

$$n=5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1+4+9+16+25=55$$

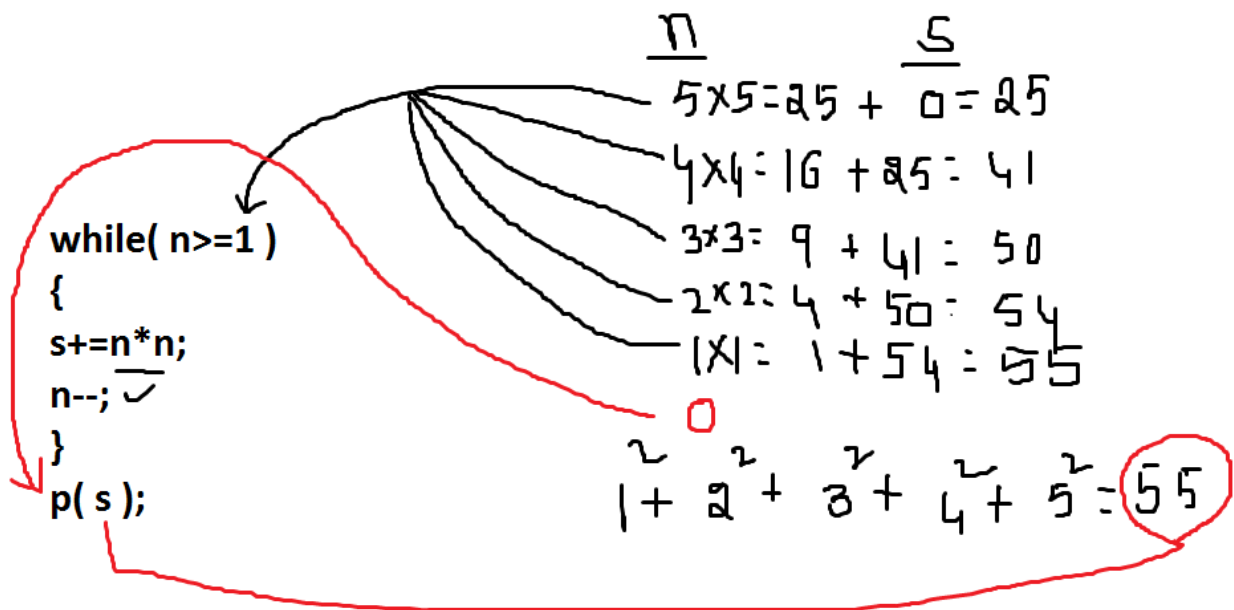
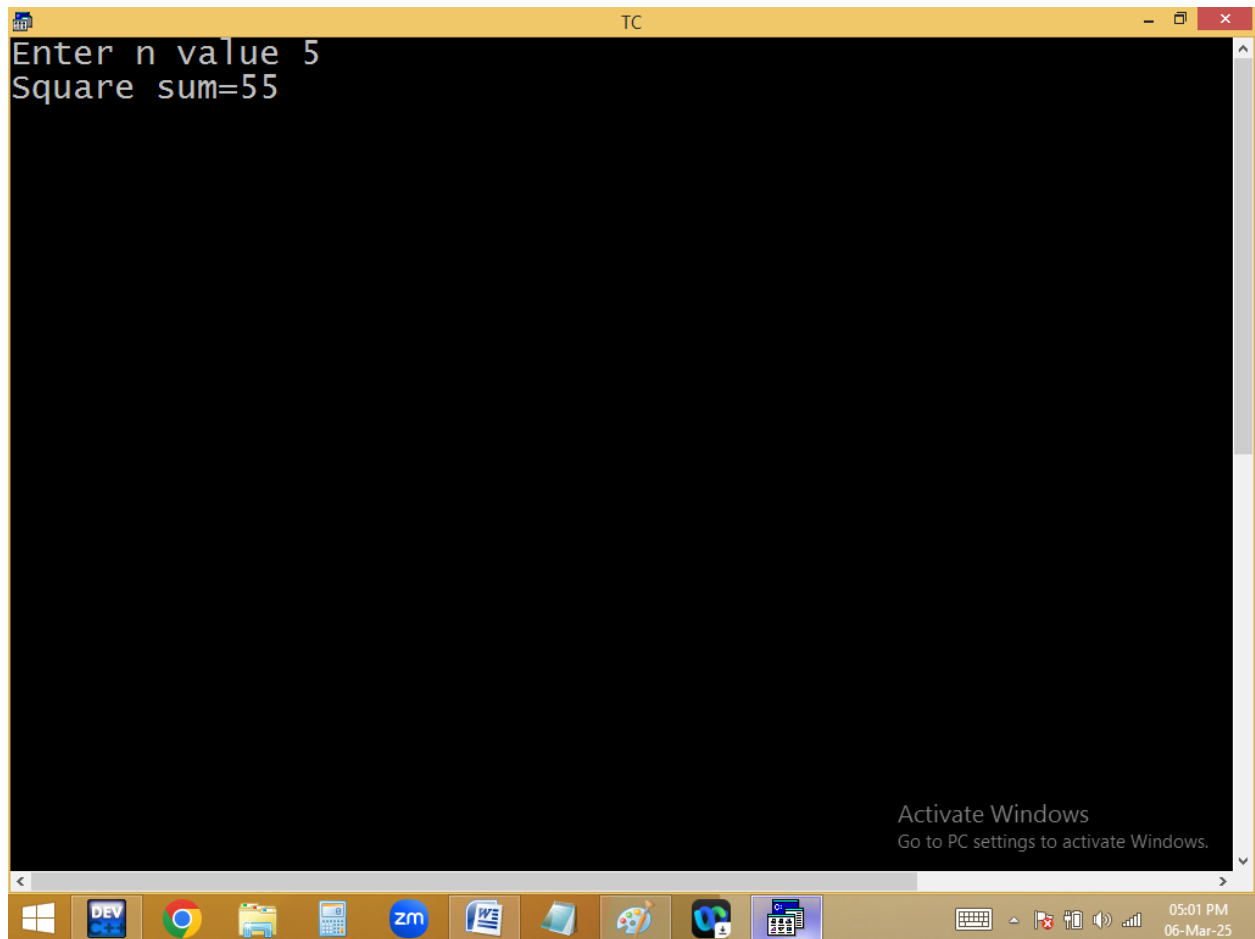


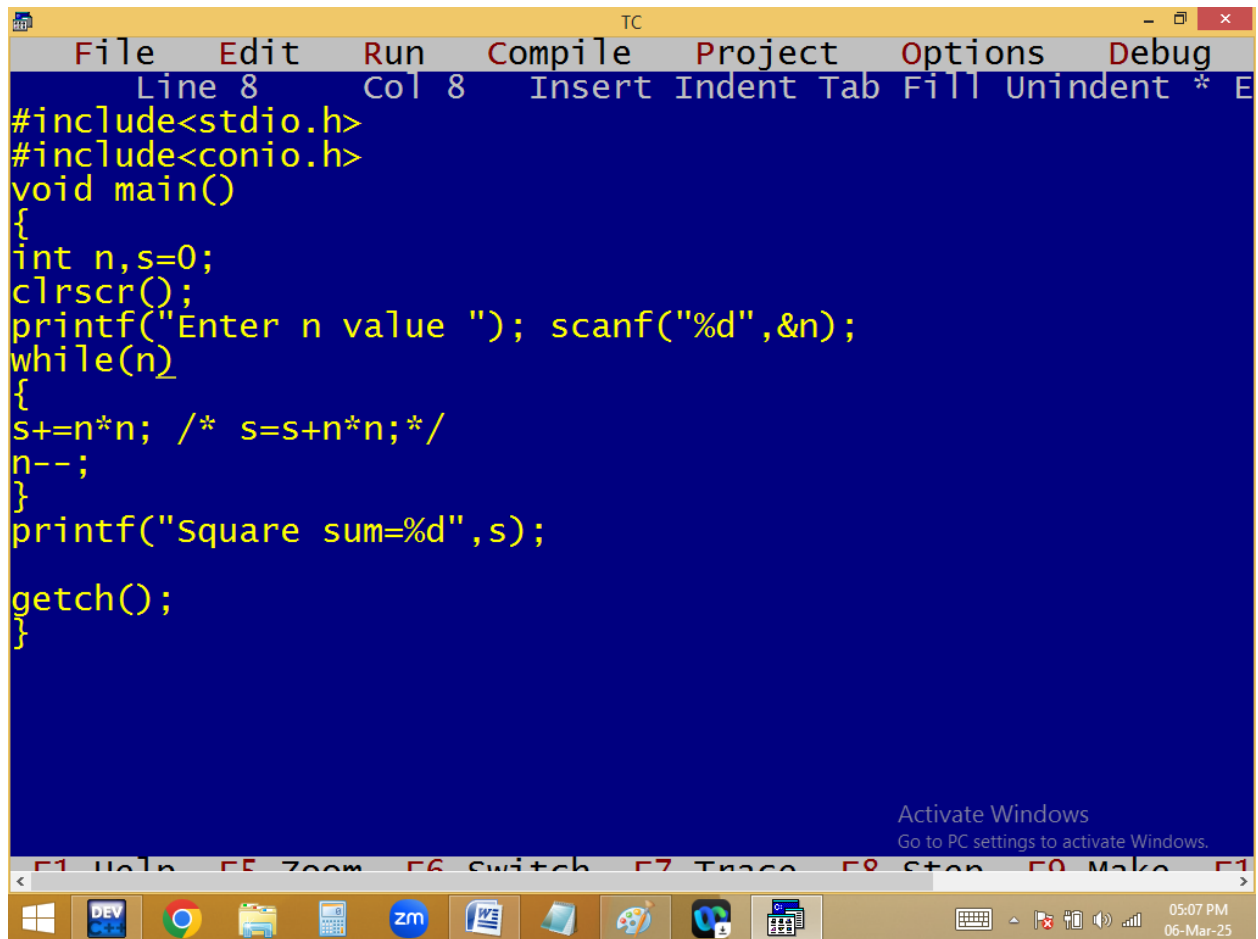
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 14 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
s+=n*n; /* s=s+n*n;*/
n--;
}
printf("Square sum=%d",s);

getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the date and time: "05:01 PM 06-Mar-25".



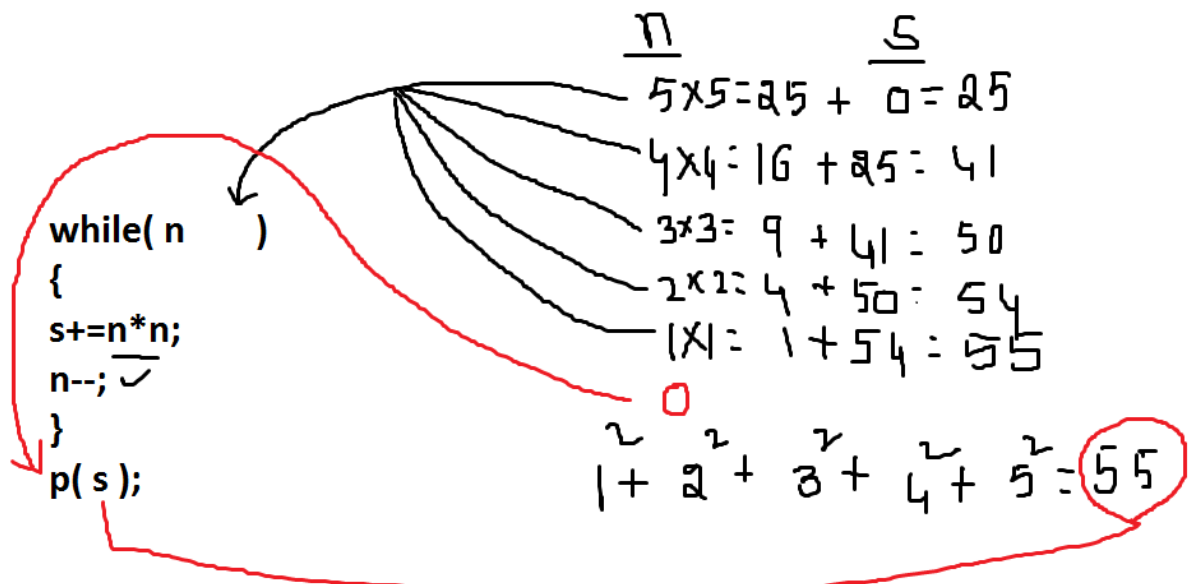
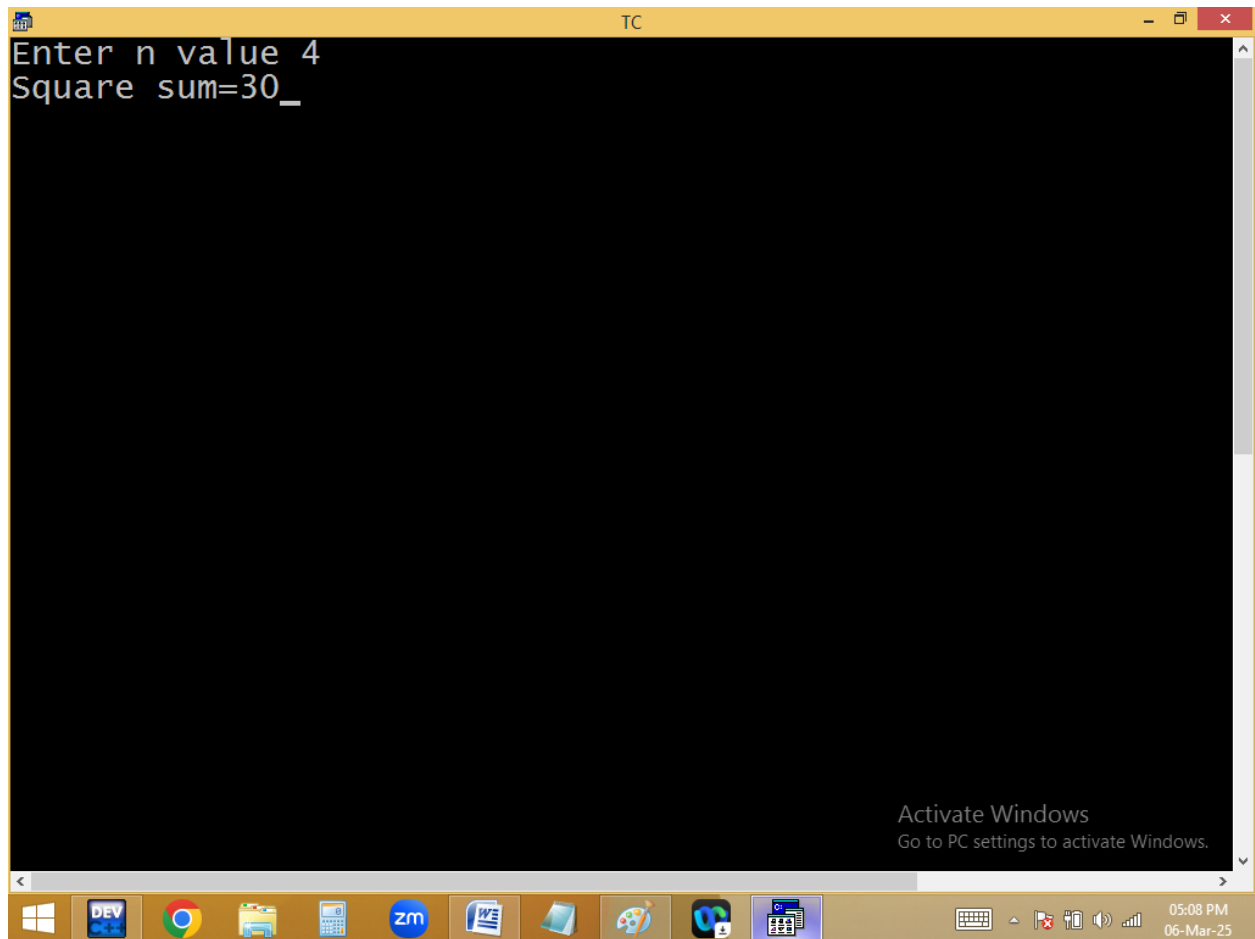


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 8", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code in yellow text:

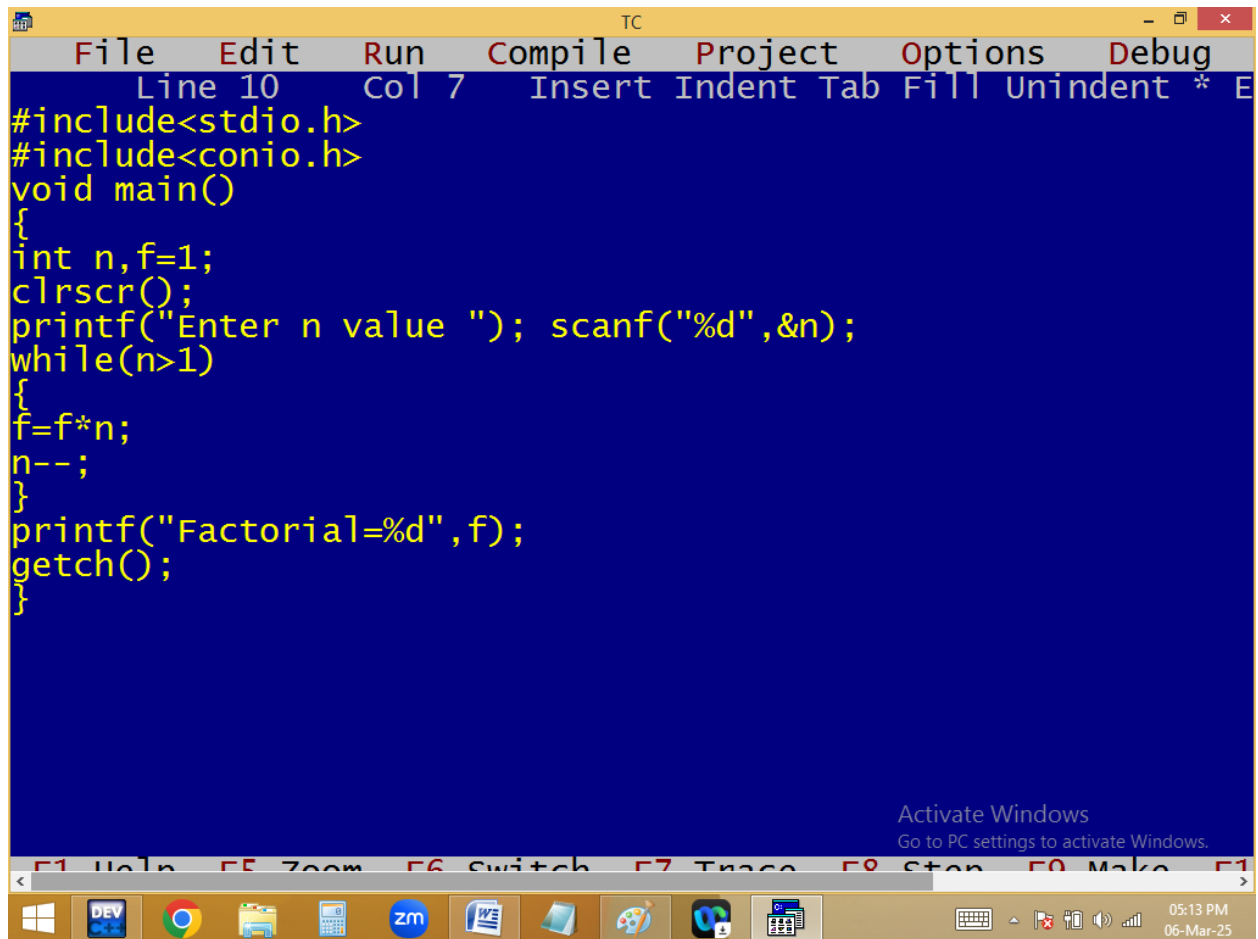
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n)
{
s+=n*n; /* s=s+n*n;*/
n--;
}
printf("Square sum=%d",s);

getch();
}
```

At the bottom of the IDE window, there is a text box with the message "Activate Windows" and "Go to PC settings to activate Windows." Below this, a toolbar contains icons for various functions, with labels F1 through F11. The Windows taskbar is visible at the very bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system clock in the bottom right corner displays "05:07 PM" and "06-Mar-25".



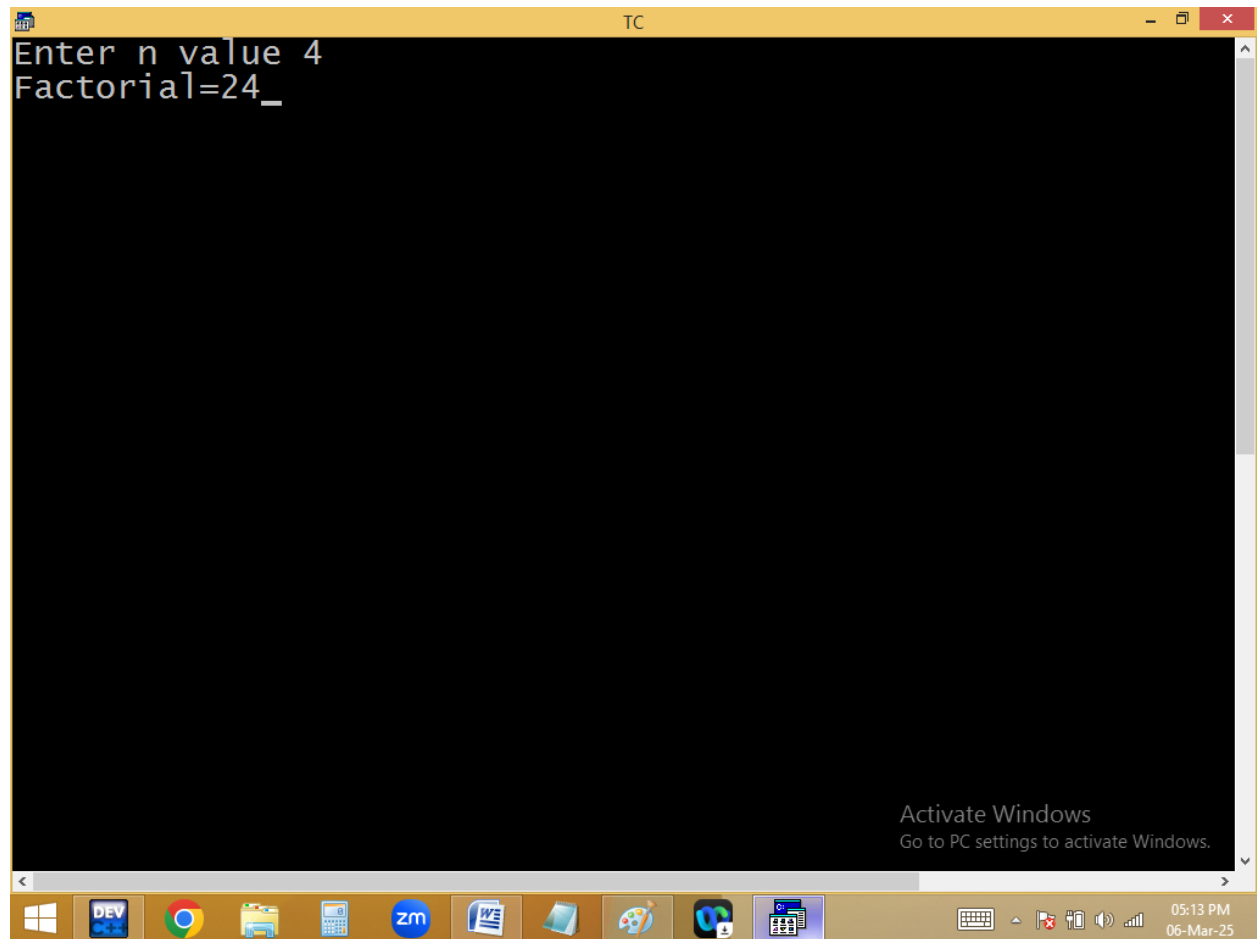
Find the factorial of given no?

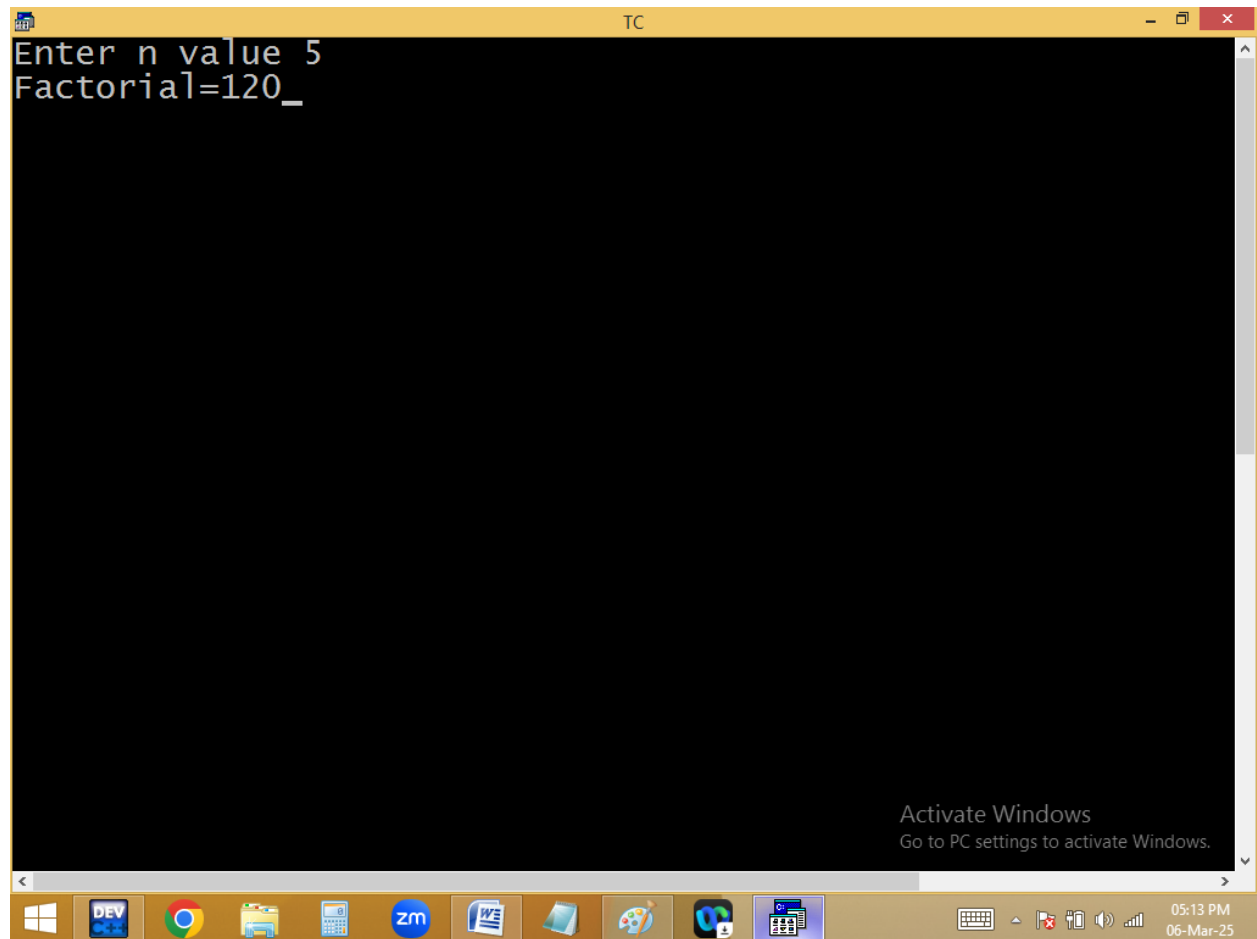


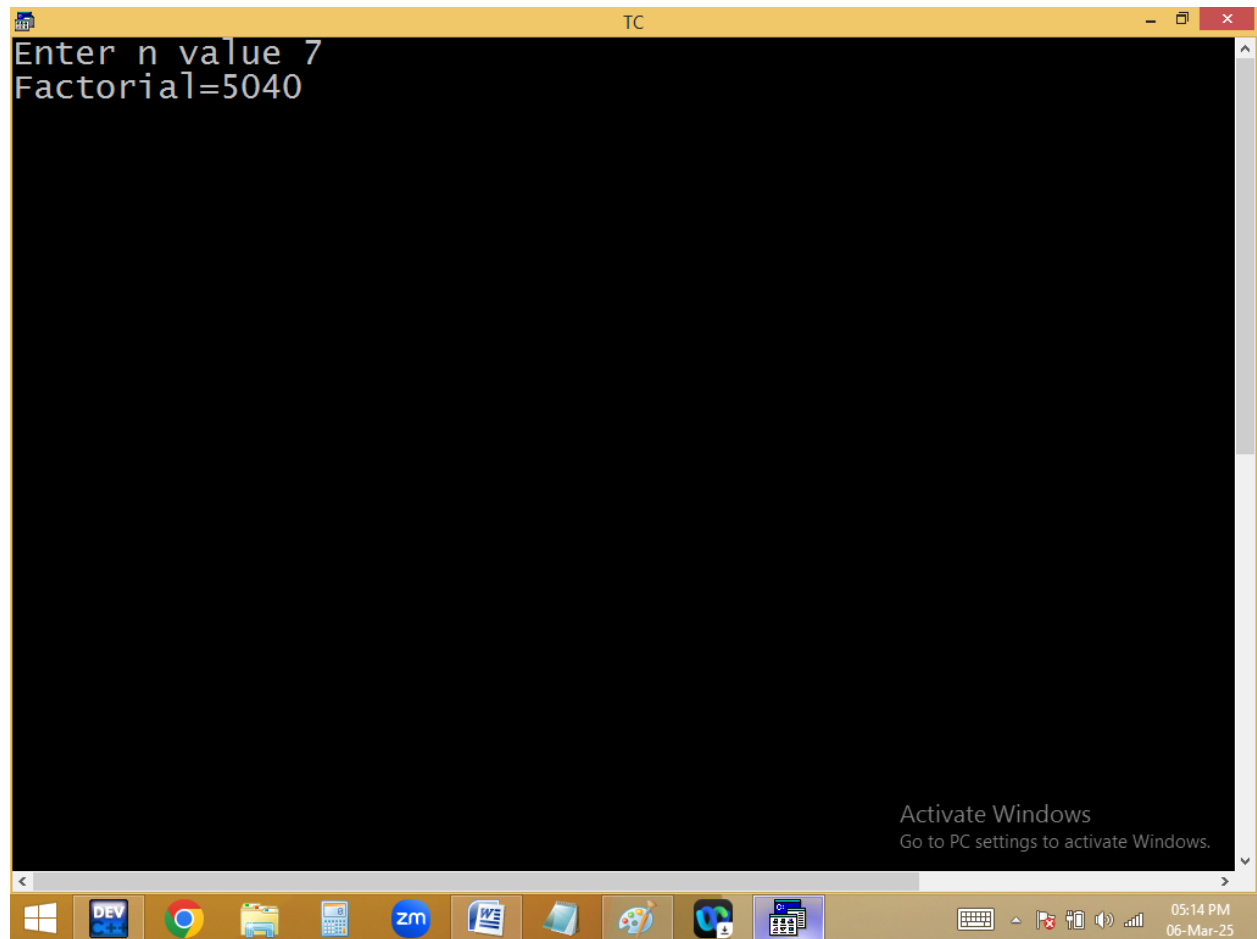
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 7" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code for calculating a factorial:

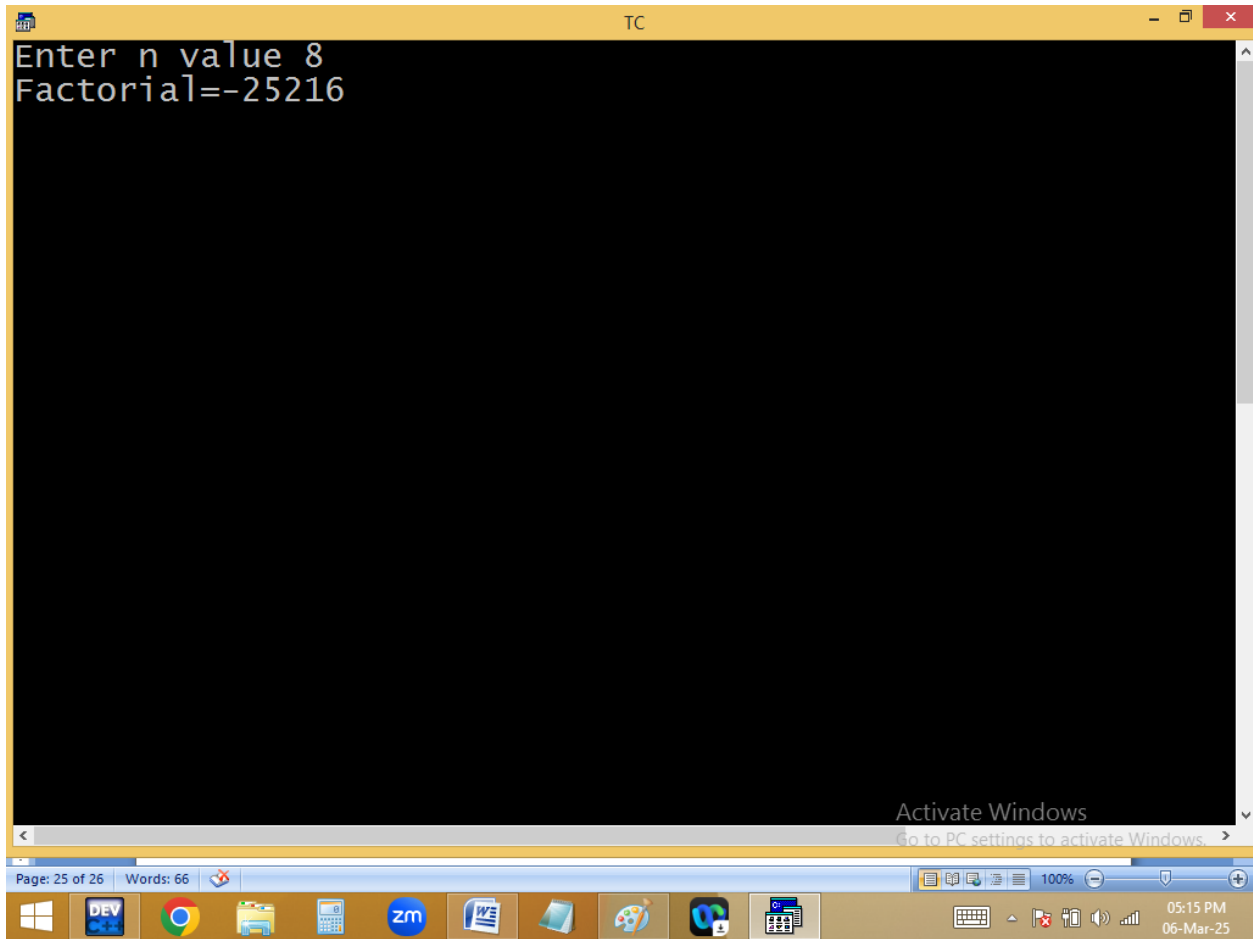
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,f=1;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    while(n>1)
    {
        f=f*n;
        n--;
    }
    printf("Factorial=%d",f);
    getch();
}
```

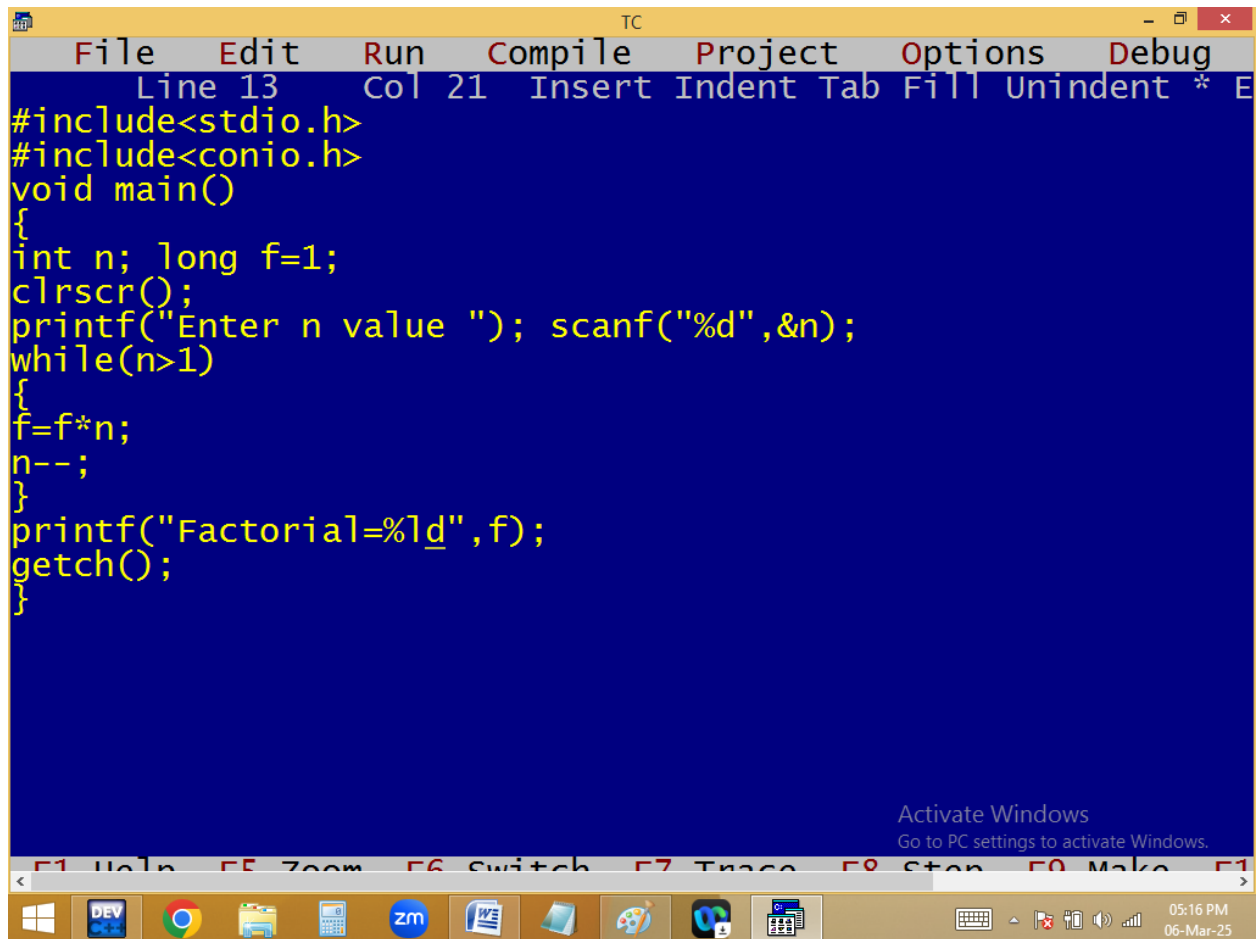
At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as 05:13 PM and the date as 06-Mar-25.











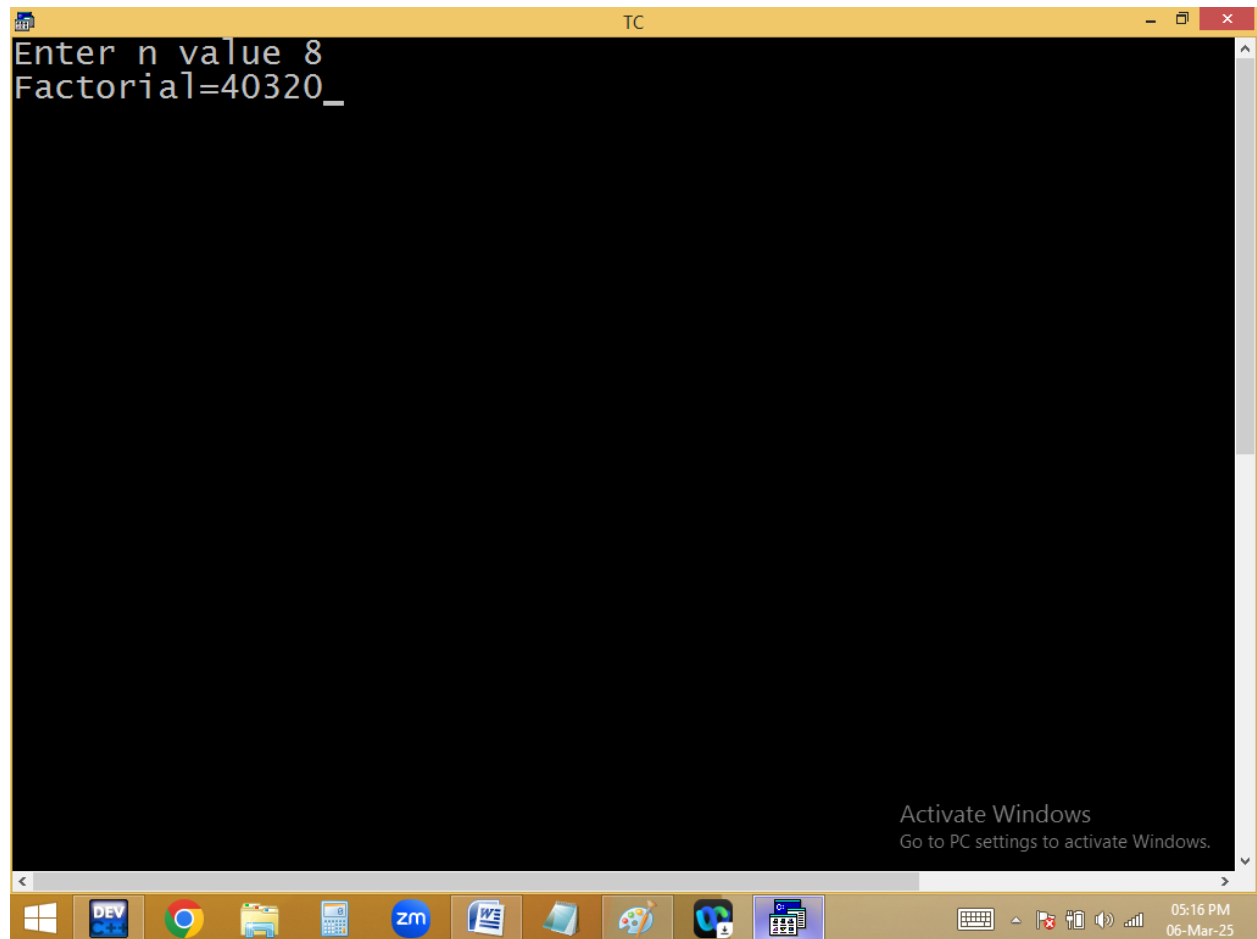
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 21", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is a C program to calculate the factorial of a number 'n'. It includes the necessary headers, defines the main function, declares variables 'n' and 'f', and uses a while loop to calculate the factorial. The output is printed using printf, and the program waits for a key press before exiting. At the bottom of the IDE window, there is a text box that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which contains icons for various applications including a file explorer, a calculator, a web browser, and a terminal. The system clock in the bottom right corner shows the time as 05:16 PM on 06-Mar-25.

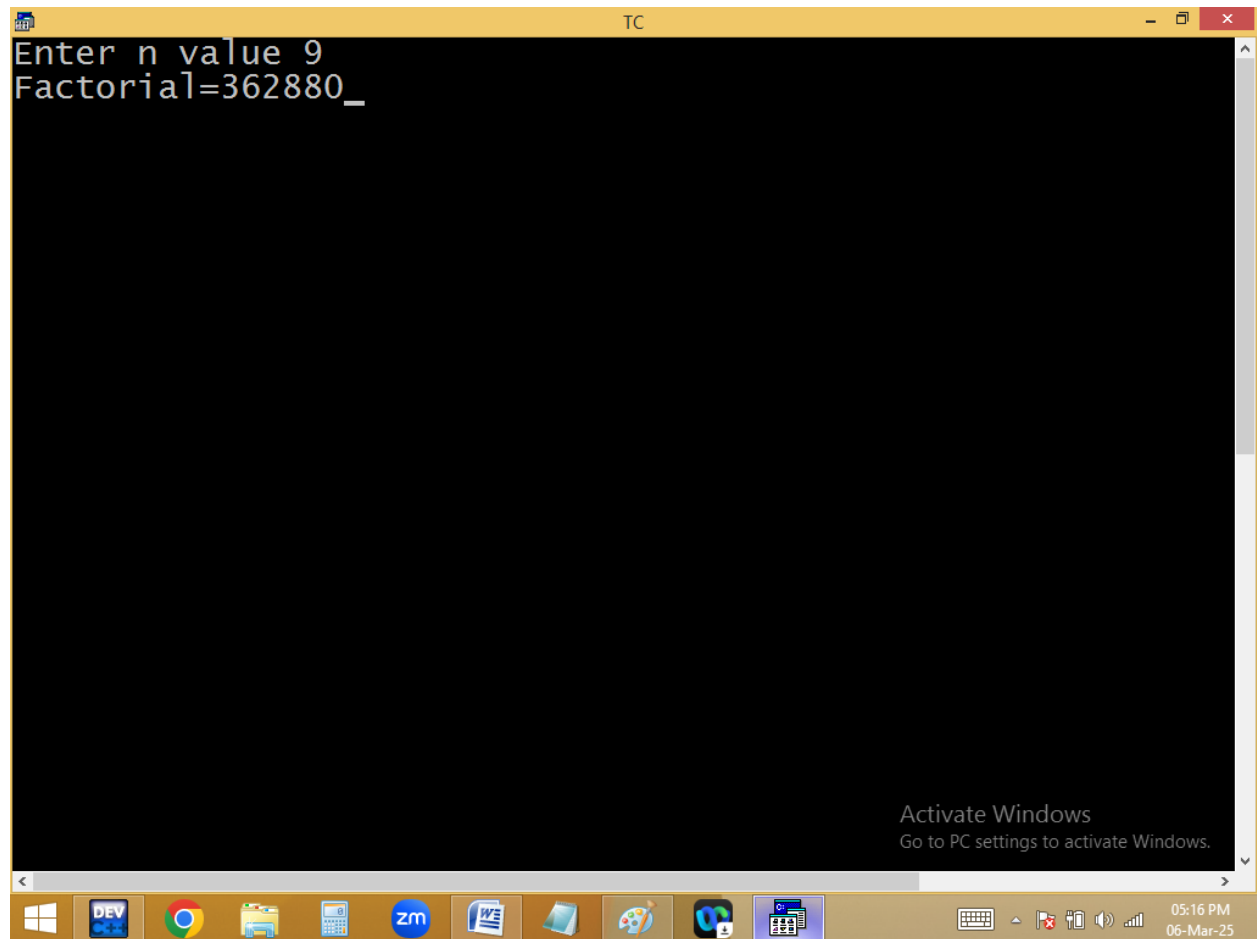
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n; long f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial=%ld",f);
getch();
}
```

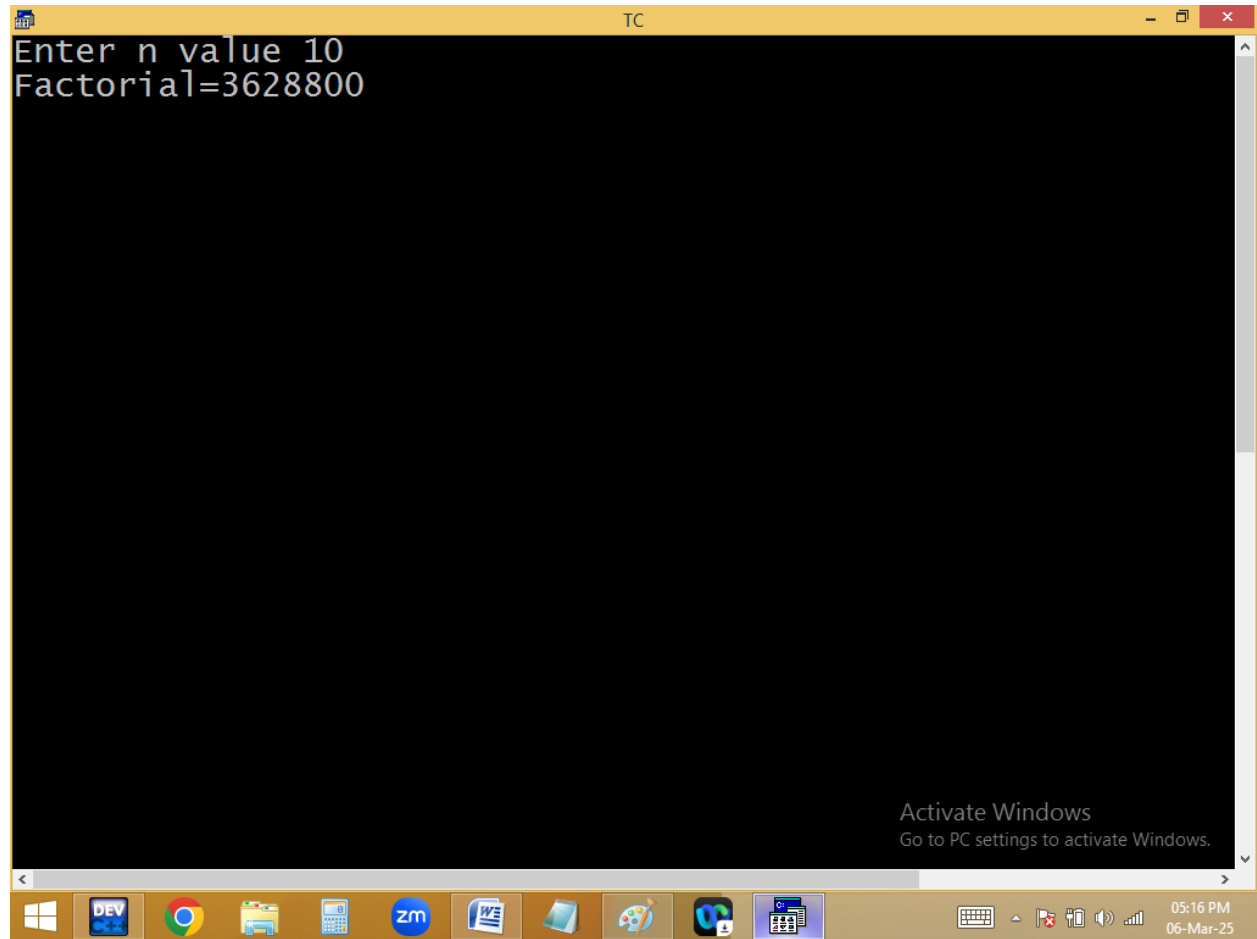
Activate Windows
Go to PC settings to activate Windows.

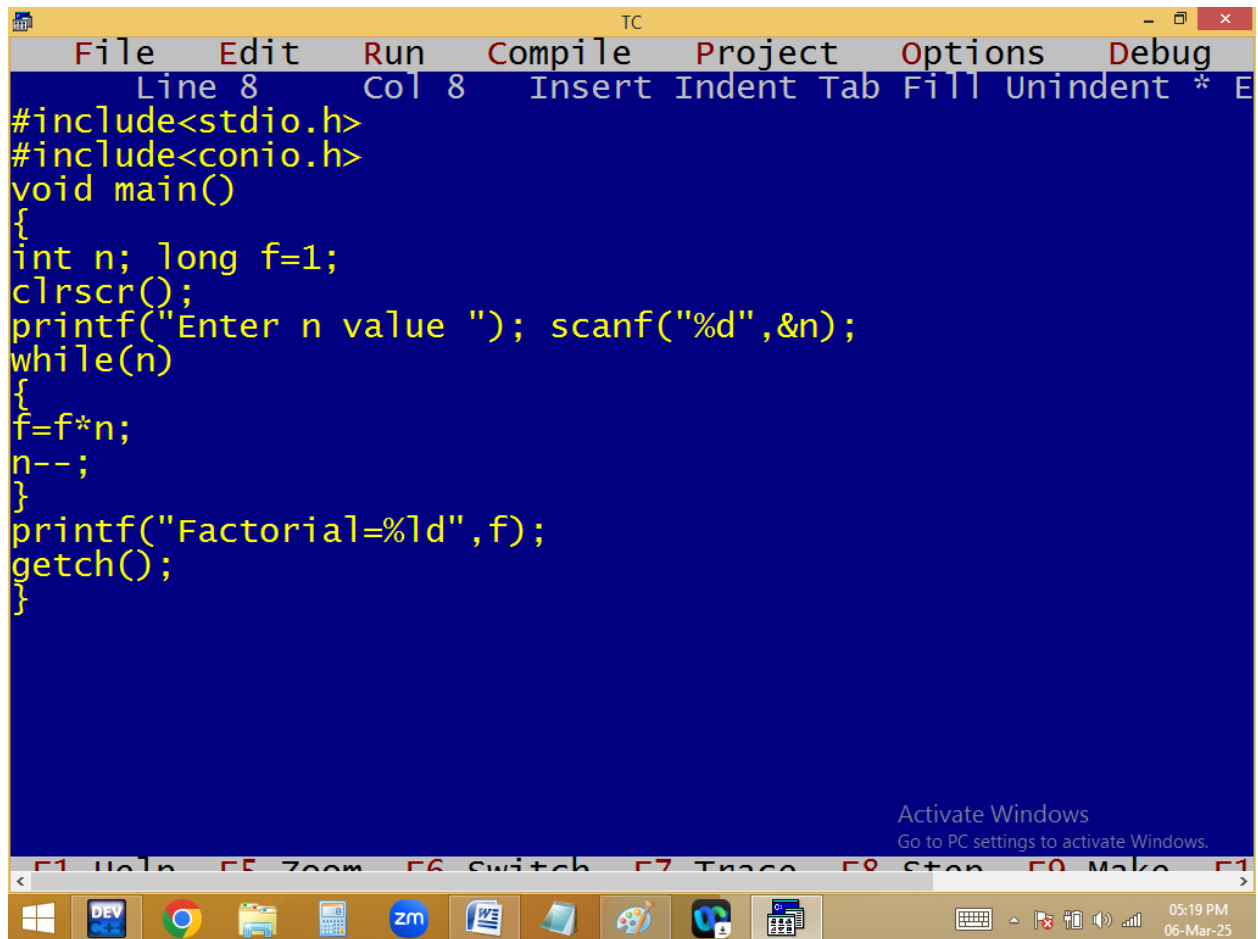
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:16 PM
06-Mar-25





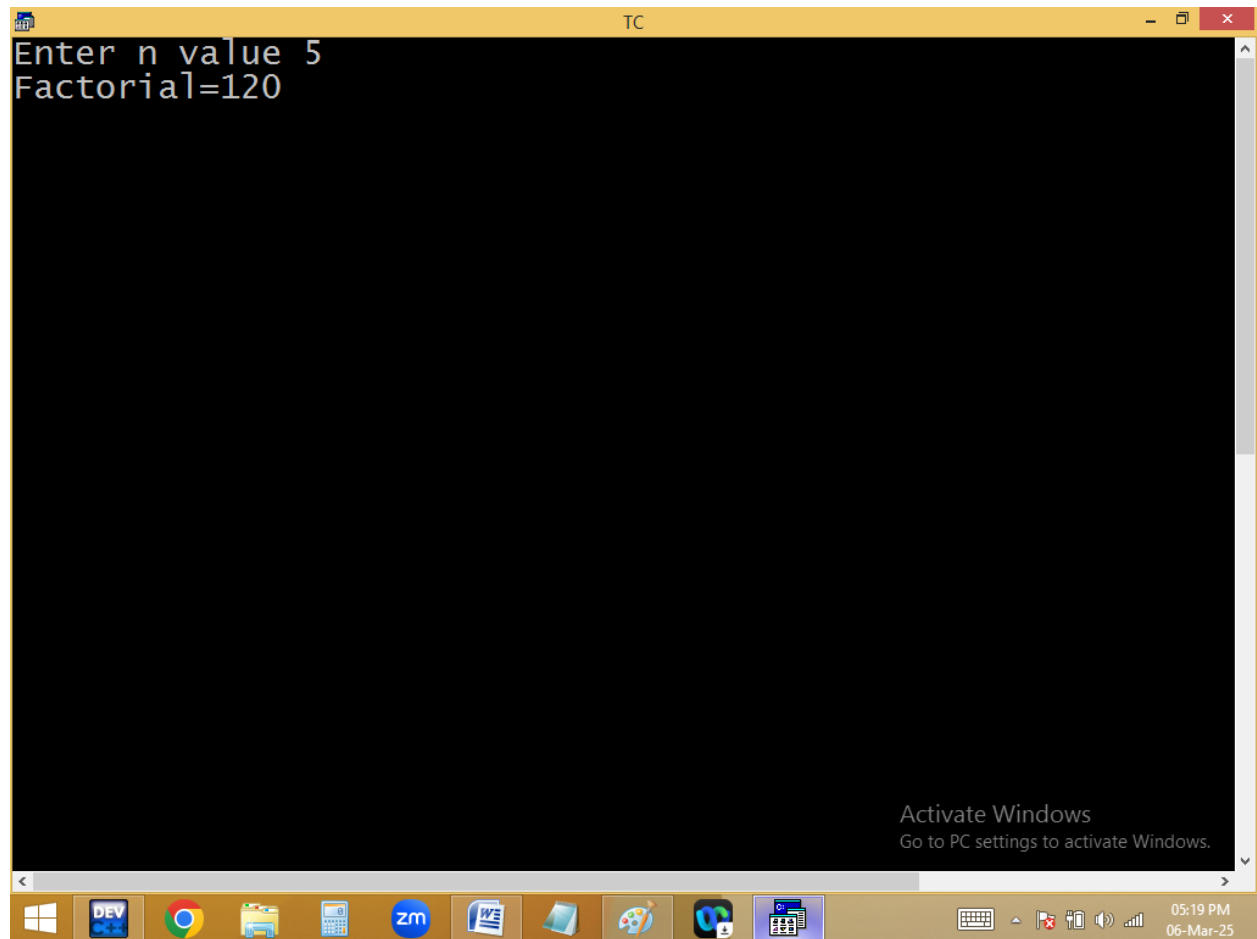


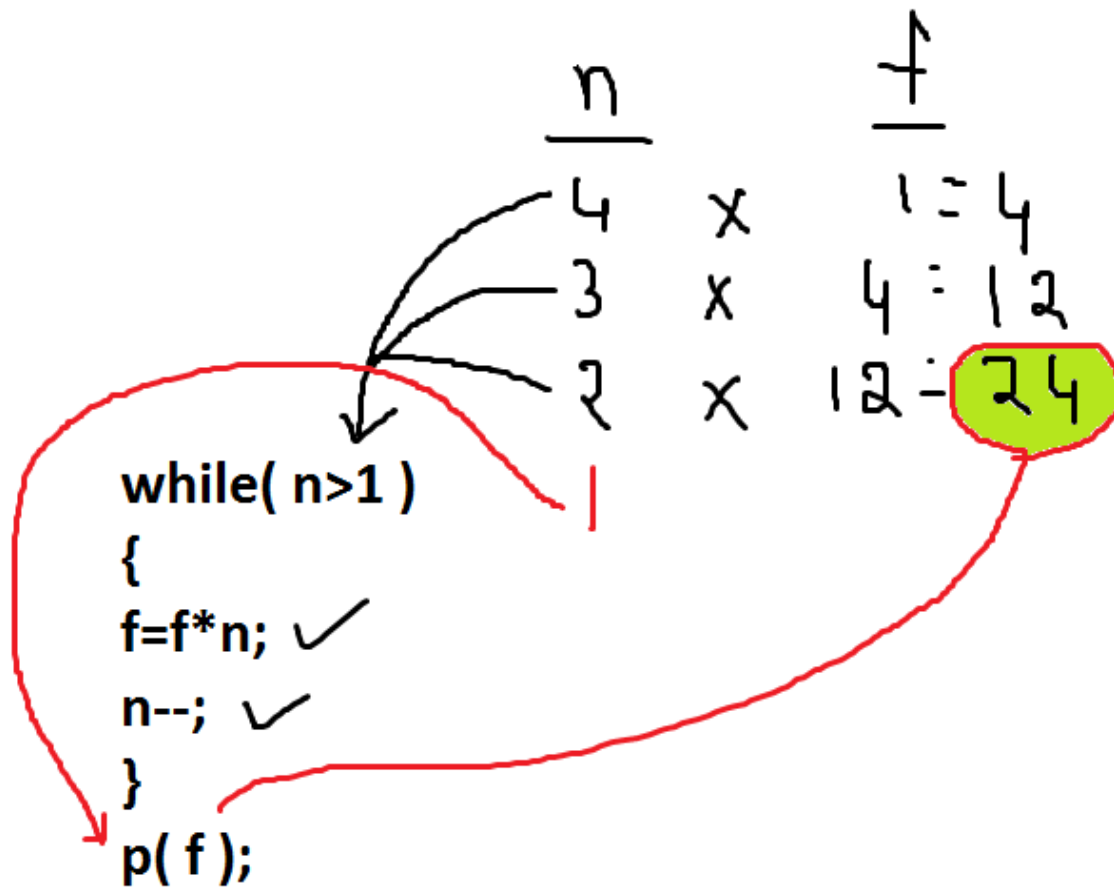


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 8", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code for calculating a factorial:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n; long f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n)
{
f=f*n;
n--;
}
printf("Factorial=%ld",f);
getch();
}
```

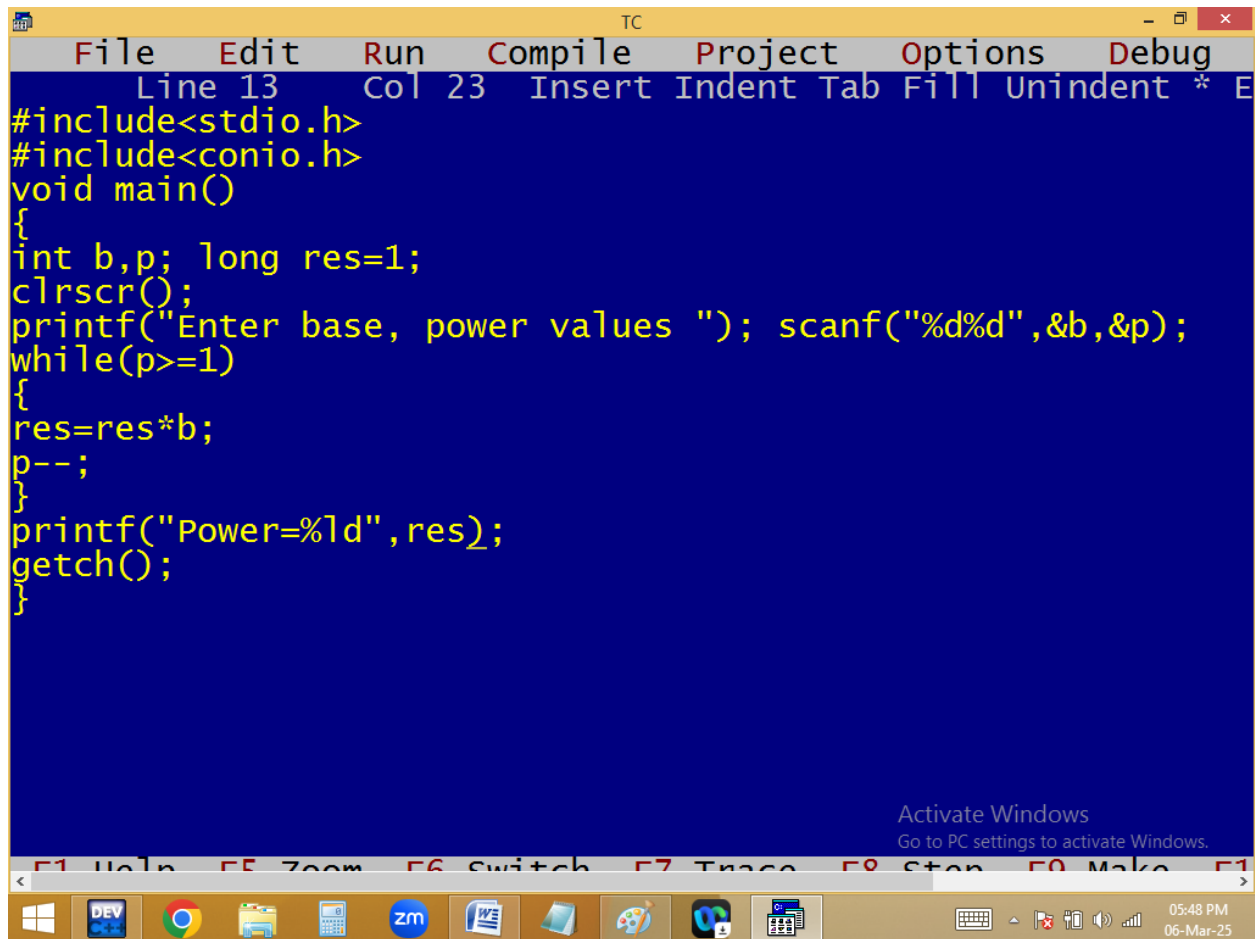
At the bottom of the window, there is a taskbar with icons for Windows, DEV C++, Google Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as 05:19 PM and the date as 06-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





Finding power using user defined program?

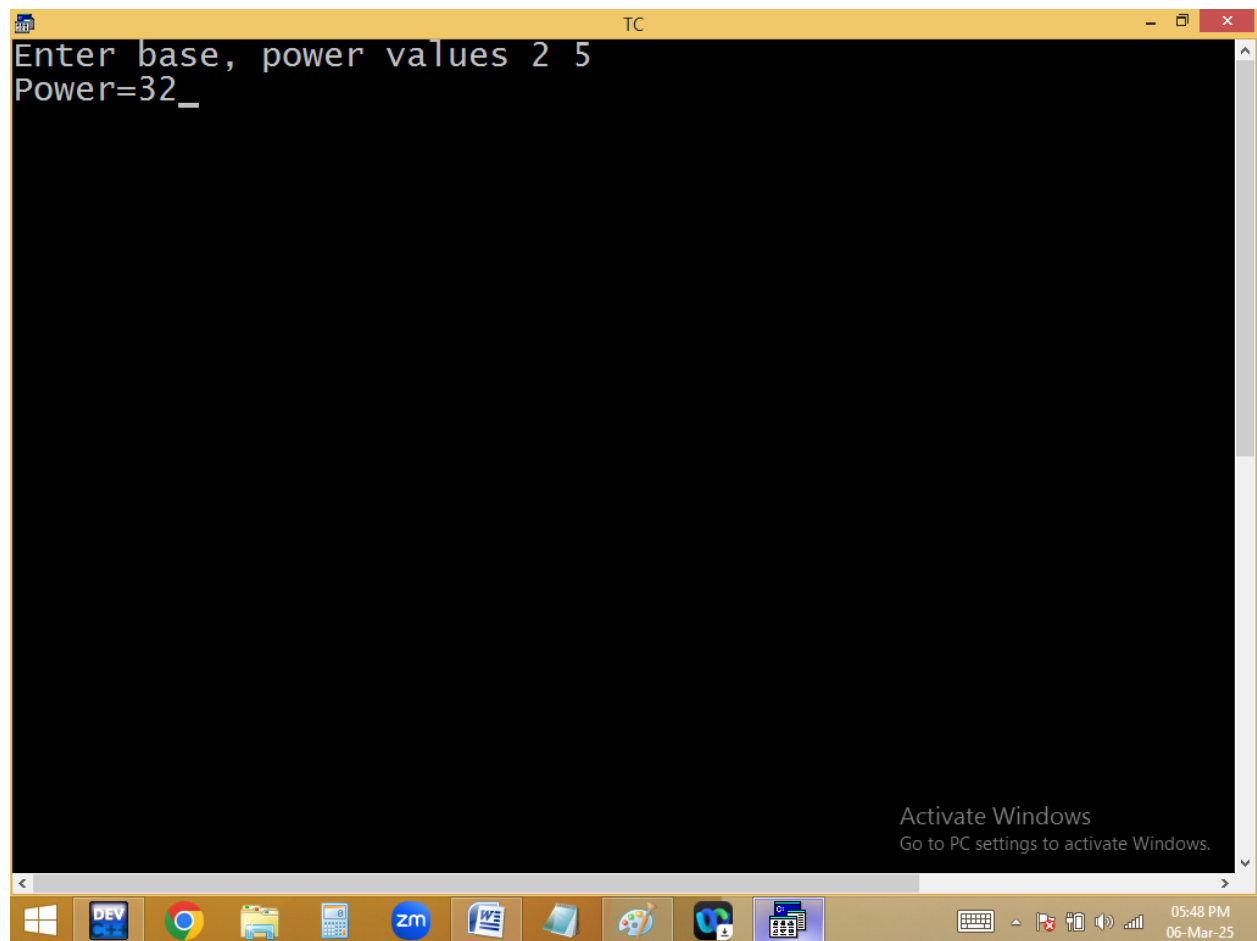
$$2^5 = 2 * 2 * 2 * 2 * 2 = 32$$

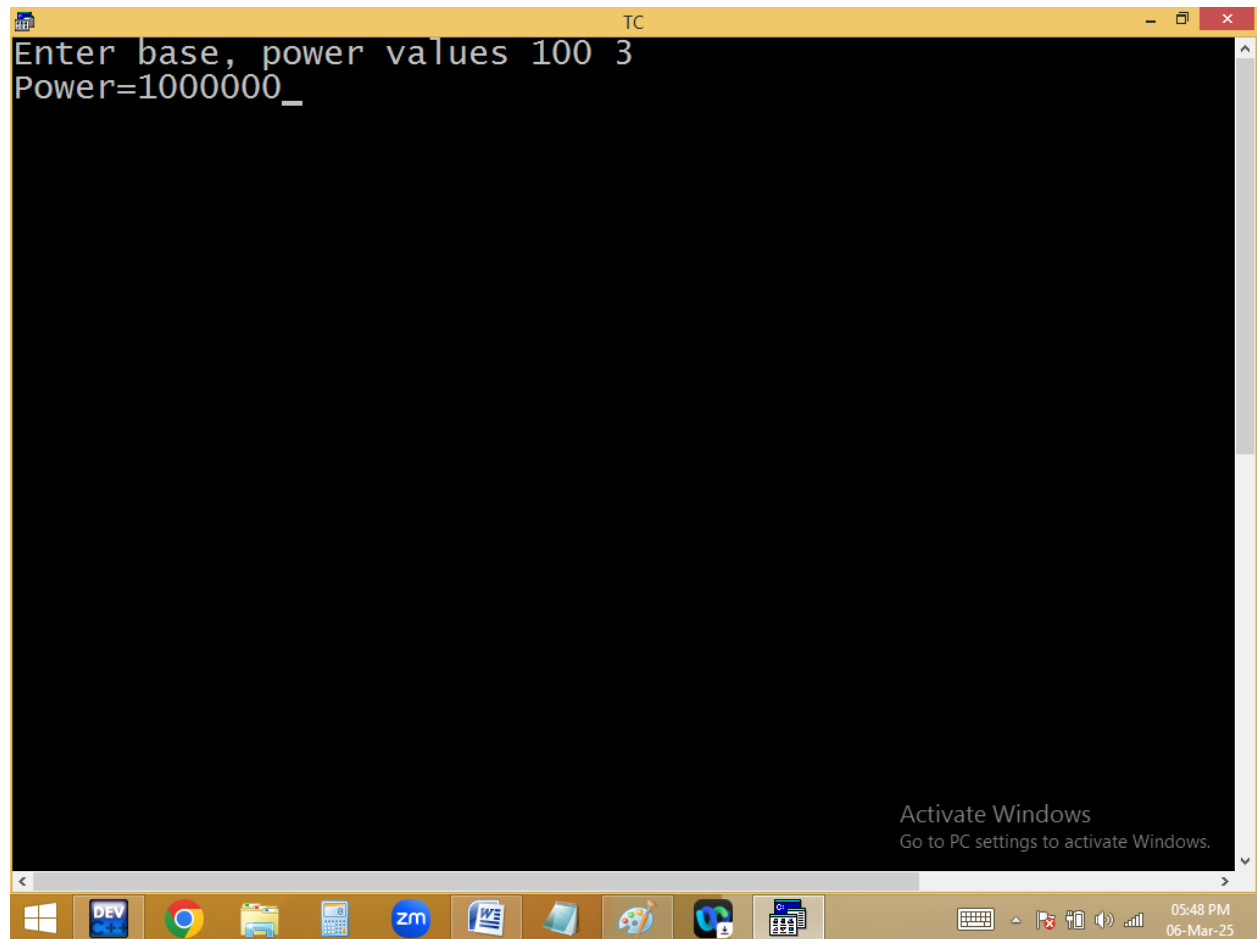


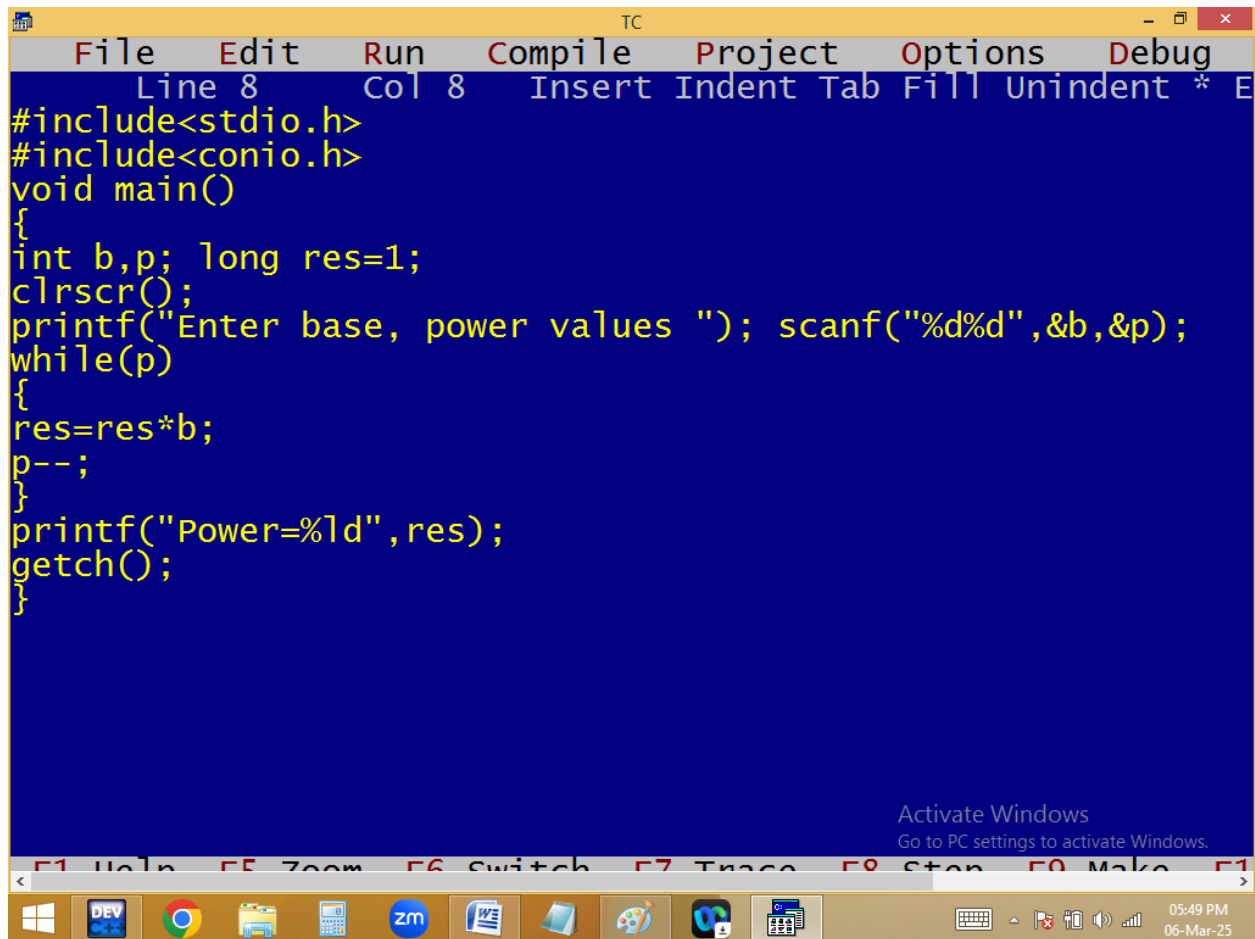
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, a submenu is open showing: Line 13, Col 23, Insert, Indent, Tab, Fill, Unindent, *, and End. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p; long res=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
res=res*b;
p--;
}
printf("Power=%ld",res);
getch();
}
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the IDE window is the Windows taskbar. It contains icons for Windows, DEV C++, Google Chrome, File Explorer, Calculator, Zoom, Word, a folder icon, a game controller icon, and a calendar icon. The system tray on the right shows the date and time: 05:48 PM, 06-Mar-25. A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.



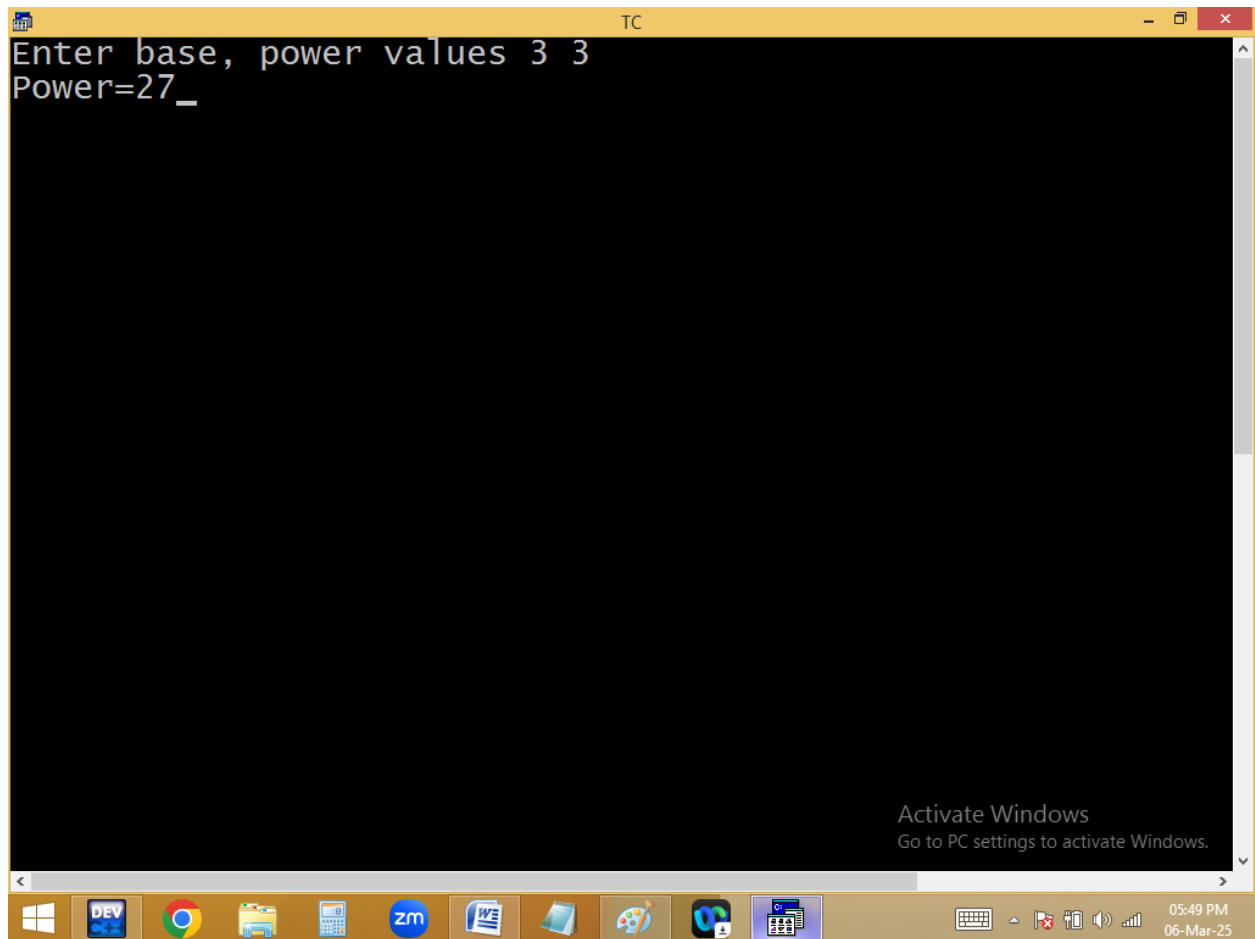




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 8", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code in yellow text:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p; long res=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p)
{
res=res*b;
p--;
}
printf("Power=%ld",res);
getch();
}
```

At the bottom of the IDE window, there is a text box that says "Activate Windows" and "Go to PC settings to activate Windows." Below this, a row of function key shortcuts is visible: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar is at the very bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system clock in the bottom right corner shows "05:49 PM" and "06-Mar-25".



The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text reads: "Enter base, power values 3 3" followed by "Power=27_". The window title bar says "TC". At the bottom, there is a Windows taskbar with various icons including DEV, Chrome, File Explorer, Calculator, Zoom, Word, and others. The system clock shows 05:49 PM on 06-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the TC window.

```
Enter base, power values 3 3
Power=27_
```

Home work →

$$1.2^{-3} \rightarrow 0.125$$

2. Finding powers sum

$$2^3 \rightarrow 2^1 + 2^2 + 2^3 \rightarrow 2 + 4 + 8 = 14$$

$$5^3 \rightarrow 5^1 + 5^2 + 5^3 \rightarrow 5 + 25 + 125 = 155$$

